Routing Is Least Learnable Where It Is Most Valuable: Bounds on Representation Routing for Web Agents

Anonymous Author(s)
Affiliation
Address
email

Abstract

Web agents observe a browser through text, pixels, or both, and the choice is usually fixed once for all tasks. We measure six observation modes across eight site–model combinations (cells) on VisualWebArena and WebArena — four of them screenshot-free, one pair differing only in the image — and ask what choosing per task would buy. The modes are complementary: each solves tasks the others miss, they fail in structurally different ways, and the best choice reverses between task sets. The obvious prize, an oracle that picks a winning mode for every task, looks large but is inflated by run-to-run noise: rerunning the same mode on the same tasks changes 12–14% of outcomes, so a second run of a mode already in hand gains about as much as adding a new one. What survives is a cost bound: sending only the tasks no mode solves to the cheapest mode cuts cost by 9.5–30.6% in 8 of 8 cells at unchanged success. We then test five routing policies (picking the mode, deciding when to spend on the strong mode, a zero-cost rule read off the task text, a confidence cascade, and pooled cost tiers), and none robustly beats simply fixing one well-chosen mode; the one exception is a fragile result in our sparsest cell. The central obstruction is that routing supervision is produced at the agent's success rate: the weaker the agent, the fewer labels a router gets, exactly where routing would be most valuable. This limit belongs to today's agents rather than to routing itself. Label supply and routing opportunity rise together (correlation 0.95 across cells), so a stronger agent can overturn the result. We report the rerun bands and the resolution threshold they imply, so the same correction can be applied to any multi-arm ceiling claim.

1 Introduction

A web agent has to be shown the page before it can act on it. In practice it is shown one of three things: the accessibility tree as text, a screenshot as pixels, or a set-of-marks fusion in which the screenshot is annotated with the tree's element ids [Yang et al., 2023]. Which of these an agent receives is the grounding decision that Zheng et al. [2024] identify as the bottleneck for a capable backbone, and that a line of work on visual grounding for GUI agents attacks directly [Gou et al., 2025]. Benchmarks such as VisualWebArena [Koh et al., 2024], WebArena [Zhou et al., 2024] and Mind2Web [Deng et al., 2023] report agents that make this choice once, at build time, and hold it for every task. The choice is treated as a design decision about the system rather than as a decision about the task in front of it.

It need not be. We hold one scaffold, one prompt budget and one action space fixed, and compare the six representations across eight (site $\times$ backbone) cells. They are not redundant: 44 of the 48 mode–cell pairs solve at least one task no other mode in that cell solved, and every mode does so in some cell. Where exactly one channel succeeds, the losing channel fails in a structurally different

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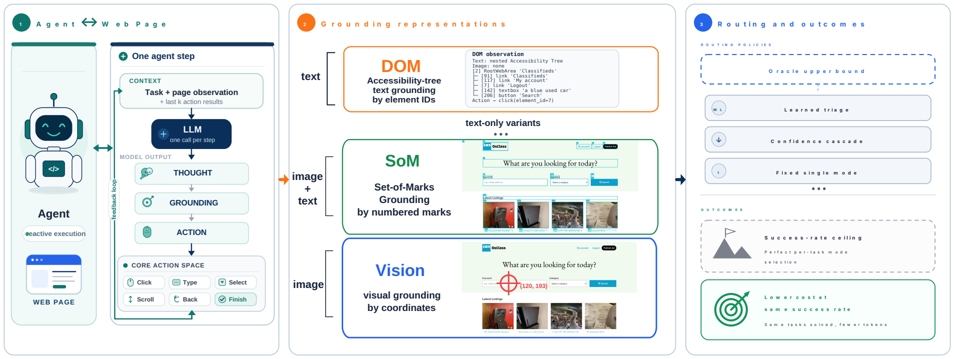

Figure 1: What this paper measures. **(1)** One model call per step emits a thought, a grounding and an action, with the last eight action outcomes as context. **(2)** That step is grounded by element id, by numbered mark, or by coordinate; screenshot-free variants separate the text payload from the prompt family (§2). **(3)** Three of our five routing policies, bracketed by an outcome oracle above and a fixed single mode below (the pooled-tier router and the zero-token rule are omitted for space; §6), and the two ceilings a perfect per-task choice could reach. Only the cost ceiling survives the rerun control of §4, because it adds no arm. **Caution:** the policy marked as exceeding a fixed single mode does so on a lossless-saving criterion under a label-shuffle null, and the one setting surviving correction is `red`·B2, whose triage signal is below chance (AUROC 0.483; Appendix Table 14). Under the stricter test no cell Pareto-dominates always-cheapest (Appendix Table 30).

way. *Which* representation wins also reverses between our two task sets, so the winner is a property of the deployment and not of the modality (§3). A per-task choice therefore has something to choose between, which raises the obvious next question: what would it be worth?

That question is easy to answer badly. An outcome oracle over six modes (it sees every outcome and picks a winning mode per task) reaches 7.1–51.9% against a best single mode of 2.2–35.6%, which reads as enormous. It is also a six-arm union quoted against one arm: a task counts as solved there if *any* of six runs solved it, and an *arm* is one mode run in one cell. The comparison it needs is a rerun. Rerunning one condition changes 12–14% of task outcomes, and 22% of tasks flip in at least one of three replicated arms (both measured in two cells and assumed to transfer to the other six; Appendix B); at that rate a second copy of an arm already in hand raises the union ceiling by as much as a different representation does. The union survives as a measured bound; what does not is the attribution of its headroom to representation diversity. What holds up without adding an arm is narrower: routing only the never-solved tasks to the cheapest mode cuts cost 9.5–30.6% in every cell at unchanged success (§5).

Routing itself is not new. Systems route a query to one of several *models* to buy accuracy at lower cost [Chen et al., 2023, Ong et al., 2025], including for web and computer-use agents specifically [Li et al., 2025, Liu et al., 2026]; Moslem and Kelleher [2026] survey the area. What those route over is a capability gradient: a larger model really is more accurate, so difficulty predicts which arm to use. We ask whether the same holds when the arms are representations of one browser state under one model, where no such gradient exists by construction.

The lower bound is where the paper turns. We build five routing policies: choosing which mode to use, a learned triage of spend, a zero-token rule read off the task text, a confidence cascade, and pooled cost tiers. None yields a robust improvement over simply fixing the best or cheapest arm, the one exception being descriptive rather than confirmatory (§6). That would be uninformative if they all failed the same way, and they do not. One dies of label supply. One has ample labels and buys nothing. One has a perfect free signal and still loses, because the expensive arm does not hurt on the tasks the signal excludes (§6).

The gap between those bounds is what this paper can offer without overstating it. What it comes down to is a circularity we call the **supply–value coupling**: routing supervision is produced at the success rate, so routing is least learnable exactly where it would be most valuable. Those look like two obstructions and are one: the labels a router can learn from and the tasks where routing can pay both track the agent's success rate, which indexes the negative result to the current capability regime and says what would overturn it (§7).

Underneath all of it, every comparison is read against a measured same-condition rerun band rather than against zero (§4); Hajimiri et al. [2026] make the same move on a different axis, showing that a token-matched vanilla baseline erases the reported gains of several agent memory and skill modules. Every quantity is read from its producing artefact at render time (Appendix A). On workloads at these success rates the resolution of the instrument is comparable to the effects being reported, and a paper that does not say so is reporting its instrument.

Reading this as a statement about visual grounding. Four of the six modes carry no screenshot, and two of those four replace the accessibility tree with the flattened marks that annotate one (§2). Only one pair isolates the pixels while holding text payload, prompt family and element-id regime fixed; every other image-on/image-off contrast moves the identifier regime with it, so this design bounds what the visual channel is worth rather than cleanly ablating it. What the wider set does establish is that *which* channel wins is a property of the deployment and not of the modality (§3). The part that transfers is the null: on workloads at these success rates, a grounding intervention worth fewer points than a same-condition rerun of the arm it modifies is not yet distinguishable from one (§4).

Contributions. (i) A *rerun-matched* way to price any multi-arm ceiling: the null for a union or oracle ceiling is a same-condition replicate, not zero, and we derive the resolution threshold that follows (§4). (ii) Applied here, the six representations are genuinely complementary and which one wins is a property of the deployment (§3), but most of the apparent oracle headroom is rerun variance; what survives the correction is the cost ceiling, 9.5–30.6% at unchanged success (§5). (iii) Five routing constructions all land at or below trivial fixed policies, and the obstruction is the supervision rather than the estimator: labels arrive at the agent's success rate (§6–§7).

2 Setup

We evaluate on VisualWebArena [Koh et al., 2024] and WebArena [Zhou et al., 2024].

Throughout, a *cell* is one (site $\times$ backbone) pair, the unit everything is measured in: sites `cls`/`red` are VisualWebArena classifieds/reddit and `WA` is WebArena reddit; backbones are B0 = Qwen3-VL-235B (API-served), B1 = Qwen3-VL-4B and B2 = Gemma-3-4B (both local). Eight cells exist; `WA`·B2 was never run. A *condition* is one mode run in one cell (36 in total), called an *arm* when it is an option a router could pick; *mode* and *representation* are used interchangeably.

Every number in this paper is read from its producing artefact at render time; none is typed by hand (Appendix A).

Each table's multiplicity status (confirmatory / descriptive / selection-derived) is stated in the appendix preamble.

A deliberately plain agent. One model call per step emits one JSON object: `thought` (free text; the backbones run in non-thinking mode, so this is the only reasoning the system produces), a self-reported `confidence`, an `action_type` and its arguments. The prompt is the instruction, the mode's system prompt, the last eight steps and the current observation; there is no planner, memory, reflection or retry policy above the step. The history records *outcomes* rather than reasoning: action, success, whether the page changed, resulting URL. Holding this fixed is what makes a difference between conditions a difference of representation and not of scaffold.

Six observation modes: a 2×2 plus two corners. Four modes carry no screenshot and differ only in which *text payload* the agent receives and which *prompt family* tells it what to expect (Figure 1, panel 2). The payload is either the full accessibility tree or the flattened `[SOM_MARKS]` list that annotates a set-of-marks screenshot; crossing the two gives DOM, DOM+sprompt (tree under the SoM prompt), DOM+stext (marks under the DOM prompt) and SoM-image (marks under the SoM prompt and, despite the name, no screenshot). The corners add or remove the image: SoM is SoM-image plus the annotated screenshot, Vision is the raw screenshot with an *empty* payload. The payload also fixes the element-id regime and therefore which modes inherit id churn (§4): `[SOM_MARKS]` is keyed $1 \dots K$ by position, the tree keeps native browser node ids. A Vision episode whose screenshot failed to capture aborts rather than degrading to text, so its "screenshot only" contract is auditable rather than assumed. The full mode grid, with per-mode payload and prompt detail, is Appendix K.

Deployment classes. We group the six into the three shapes web agents are deployed in, *no-image* (DOM and the three screenshot-free variants), *hybrid* (SoM) and *vision-only* (Vision), then compare at one arm per class, since a maximum over the four no-image arms against one is not a comparison. The grouping is a deployment convention rather than a claim the data licenses: the behavioural evidence separates Vision and SoM from the rest but cannot pool the four text modes (Appendix Table 5). Appendix D has the full comparison and the ablation (Table 4).

Task universes. Rates are formed over a *scored* set of 224 classifieds tasks, 203 reddit, and the 104 WebArena tasks all six modes ran. That set is what we collected minus protocol exclusions. We wrote rules about task configuration rather than picking tasks, so how many each rule removes is its output, not our choice. Classifieds loses none; reddit loses two of 205. One is excluded on its configuration alone: its evaluator only asserts that three subreddits are *absent* from the sidebar and never checks the subscription the task asks for, so doing nothing scores 1. The other is a cross-site task the model can answer from memory: it accounts for 9 of the 11 cross-site successes on reddit, and none of those 9 ever loaded the Wikipedia host the task points to. Every artefact records which task set it was built on, so an analysis using the older one fails a check rather than silently mixing denominators. Both exclusions are reported with and without.

Binary scoring. Over 7,686 scored episodes in 36 conditions the evaluator emits exactly two values, none missing and none non-numeric (Appendix Table 41). There is no graded target: a partly completed task is indistinguishable from one never begun. That is a property of the benchmarks rather than of this pipeline, and a precondition of every routing negative below.

Efficiency estimands. An *estimand* is what a number is actually measuring, as opposed to what it is called. Cost here is per-episode billed cost, but B0's per-token API bill and B1/B2's electricity estimate are not the same kind of quantity, so they can be ordered and not divided. Within a cell, mode-to-mode ratios can therefore be read as ratios; across deployment classes only the ordering carries; the two are never pooled. Switching the denominator from per-attempt to per-success changes which mode is cheapest in four of the six cells with enough successes to divide by. A latency figure is mostly environment: the model call is 22–67% of a step. No bare cross-site latency number appears in this paper (Appendix G, Tables 12 and 13; the per-mode efficiency matrix is Table 9).

3 Different representations fail differently

Unique solves. The case that routing could be worth anything starts here: the modes are not redundant. Two prior results bound what this can claim. Xie et al. [2024] report the field's most-cited modality comparison (tree, screenshot, both, set-of-marks), but as an ablation inside a benchmark release rather than a controlled study, and Levy et al. [2026] *engineer* complementarity, with 80 tasks built so the answer is reachable through one modality and hidden in the other. What we add is complementarity *measured* on unmodified tasks, where nothing was hidden on purpose. 44 of the 48 mode–cell pairs solve at least one task no other mode in that cell solved, and every mode does so in some cell (the `unique` column of Table 6). On the tasks where exactly one channel (the text-only or the image-bearing half of the modes) succeeds, the losing channel's failures are not generic: the paired attribution (Table 33) and the vocabulary-free probes that confirm the residual is not a ruleset artefact (Table 32) are in Appendix I, with the text/prompt axis decomposition and the hallucinated-reference counts (Tables 35, 37). Part of the difference is structural rather than behavioural: Vision is on the coordinate dispatch path by construction, and that path delivers a click far less often than the element-id path (Table 38, Appendix J).

Divergence at the first step. The difference is present from the start: on the tasks whose outcome two modes disagree about, 72–100% of VisualWebArena trajectories have already diverged by step 3 and at most 3% diverge at step 10 or later (Appendix Table 34). Accumulated drift would share a prefix and part late, so what the representation changes is the first decision rather than a compounding error. The effect is markedly weaker on WebArena (50–90% early, up to 22% late), so the two sites

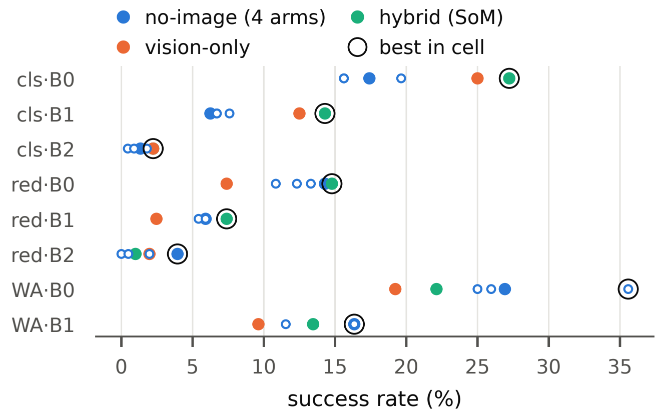

Figure 2: Success rate of all six modes in all eight cells (Appendix Table 1), coloured by deployment class; rings mark the best mode: hybrid or vision-bearing on classifieds, no-image on WebArena.

are quoted separately and never pooled. The shape holds on both, though only the VisualWebArena cells make it categorical.

Behavioural separation. The modes differ in how they act, not only in what they solve, and the difference clears the noise floor of §4: on the replicated cell, 22 of 25 live behavioural metrics separate the modes by more than a rerun of a single arm moves them, several by 5–22× (Appendix Table 11). The signatures are legible rather than statistical: on `red·B2`, Vision scrolls 8× as often as SoM and clicks half as often (Appendix E, Tables 7–8). And the divergence is not only in the macro action mix: per-step decision quality moves more than macro action frequency in 7 of 8 cells, at 1.34–4.07× on the six VisualWebArena cells with the single exception `WA·B0` at 0.97× (Appendix Table 36), so what the representation changes is which action is chosen, not merely how often each type appears. The three that do not separate are the two latency metrics (0.87×, 0.84×, which the latency decomposition reaches independently) and parse-fail rate, near zero under every mode.

No portable winner. Which representation wins is not portable either. Figure 2 plots the raw success rate of every mode in every cell (Appendix Table 1); Appendix Table 3 groups the modes into the three deployment classes of §2, at one arm per class. *hybrid* is the best class in four cells and *no-image* in three, with one tie, and which class wins reverses between task sets: hybrid leads on classifieds by 8.04–9.82pp while no-image leads on WebArena by +4.81pp at one arm per class. Those three gaps clear the rerun threshold of §4; the remaining five cells sit at 0.49–2.96pp, within noise, and are read as directions.

4 The noise floor

Appendix Table 10 is the supervision problem underneath every routing analysis in this paper: the rows a router would learn from are the contested ones, and those are exactly the rows that flip between two runs of the same condition. On `cls·B0`, rerunning one condition changes the outcome of 12.1–14.3% of tasks, and 49 of 224 tasks flip in at least one of the three replicated arms. Those flips are not spread evenly. They concentrate in the rows where more than one mode succeeds: 48–52% of those flip, against 2.9% of the rest.

What was replicated. (full tables: Appendix F) Three arms of `cls·B0` (DOM, Vision, SoM) at $n = 224$, and five modes of `wa_red·B1` on a registered ten-task draw. Nothing else: no VWA-reddit cell and no B2 cell carries a replicate, so every band below is imported into those cells rather than measured in them. On the three full arms, re-running one condition moves the set of solved tasks by 4.91–7.59pp in each direction (12.1–14.3% of tasks change outcome); on the local backbone the same quantity is 2.00–4.00pp (6.0% of tasks).

Band versus threshold. Two quantities could each be called a noise floor, and only one of them is one. The *observed band* is how much mean success rate actually moved across those pairs: 0.89–2.23pp. Each of those is *one draw* of a random quantity rather than a bound on it, so reading a 2.2pp effect against a 2.23pp "measured floor" compares a draw to a draw. The *resolution threshold* asks a different question: how large must an effect be before a single rerun would be unlikely to produce it unaided? Under the exchangeability null in which each of the d discordant tasks falls either way with probability one half, $\mathrm{SD}(\Delta\mathrm{SR}) = \sqrt{d}/n$ over the n scored tasks, giving 2.32–2.53pp, the same size as the observed band; an effect has to reach roughly 3.8–4.2pp to clear it. We use the threshold throughout, and quote the observed band only as what repetition happened to deliver. Neither is a significance test. A rerun band states the resolution of this instrument, meaning how small a difference it cannot separate. Where this paper needs an inferential claim it uses a paired interval on the contrast itself (Appendix Tables 20 and 23). The null behind the threshold models only exchangeability of the two runs, not environment drift; and site state accumulates rather than resets, so drift pushes in one direction and adds to the movement the null already allows. So 3.8–4.2pp is itself a lower bound on what repetition could deliver.

Mean differences versus set differences. Two runs of one condition also differ as *sets*: $|\{a \text{ solves}\} \setminus \{b \text{ solves}\}|$ runs 4.91–7.59pp on the three full arms. A union ceiling moves by that functional, so an added arm is judged against it in §5 rather than against the threshold above. The two must never be compared to each other. A mean-difference threshold applied to a set-difference gain would be arithmetic across estimands, which is how the same replicate can appear to license opposite conclusions.

Sources of run-to-run movement. Sampling is not the answer: every condition in this paper decodes at temperature 0 with a fixed seed. Inference is nonetheless not bit-reproducible in general [He and Thinking Machines Lab, 2025, Yuan et al., 2025], and language models are separately sensitive to formatting choices that carry no information [Sclar et al., 2024]. Two sources survive a source-level audit of the pipeline. The first is element-id churn. The accessibility-tree payload is keyed by native browser node ids that are reassigned per snapshot; the model is sensitive to those id tokens; and a replay that shuffles ids while holding the page fixed changes which element is chosen on 20.0% of B1 steps and 12.5% of B0 steps. Because churn is a property of the text payload and not of the prompt, it reaches exactly the two modes carrying an accessibility tree, DOM and DOM+sprompt. The modes carrying `[SOM_MARKS]` are keyed $1 \ldots K$ by position and are unaffected. The second applies only to B0. A hosted mixture-of-experts endpoint routes and batches server-side, and no client-side setting controls that. B1 and B2 are served locally and decode greedily, and their steps replay bit-identically (133/133 steps). Their *episodes* still move, because an episode also depends on site state, wall-clock and session lifetime. What every band here names is therefore run-to-run movement including environment drift, never decoding stochasticity. We report no split of the total across these sources. An earlier linear decomposition was withdrawn as a category error, and the per-channel figures above are what each channel moves on its own rather than shares of one budget.

Consequences for the rest of the paper. Every comparison after it is read against these bands, whether between modes, between deployment classes, or between a representation and a rerun. An arm added for its representation therefore has to be priced against an arm added by repetition (§5), and the tasks a router would be trained on are the least stable rows in the dataset (§6).

5 A rerun-corrected upper bound

Appendix Figure 3 shows the two ceilings a perfect per-task choice could reach (per-cell values in Appendix Table 16), and only one survives its own control. The *success-rate ceiling* (any mode solves it) runs 7.1–51.9% against a best single mode of 2.2–35.6%; but it is a six-arm union quoted against a one-arm baseline. Matched at one added arm, the two purchases are: **a genuinely different representation buys +1.97 to +7.14pp; a second run of an arm already in hand buys 4.91–7.59pp** wherever a replicate exists. These are the same quantity at the same arm count in the same cell, so they are directly comparable, and repetition is not the cheaper of the two by accident: it is the larger. The *cost ceiling at matched success* adds no arm and is untouched by that objection: keeping the best mode everywhere and sending only never-solved tasks to the cheapest arm leaves success unchanged by construction and cuts cost 9.5–30.6% in 8 of 8 cells. The corresponding decomposition appears in Appendix Tables 17–18.

The same rerun correction applies to fusion as a representation purchase: Appendix Table 19 puts a distinct arm and a rerun side by side at the same arm count, and against the workload-matched single channel the fused mode's premium clears the rerun threshold of §4 in 0 of 8 cells (Appendix Figure 5 and Table 20).

The number worth aiming at is therefore not the six-arm union. It is the gap between the fixed policies of §6 and the *cost* ceiling: 9.5–30.6% in 8 of 8 cells at unchanged success, reachable in principle with one bit per task rather than a mode identity. That target survives the rerun objection for the reason the success-rate ceiling does not: it adds no arm, being the same arms spent differently.

6 The lower bound: five routing constructions, five obstructions

Five constructions, five different obstructions, one outcome: none beats simply fixing the best or cheapest arm (per-policy tables in Appendix H). That the obstructions differ is what makes the bound informative rather than an artefact of one weak router. A single exception is descriptive rather than confirmatory: on `red·B2`, our sparsest cell, the zero-token rule exceeds always-DOM on success, cost and latency at once (Appendix Table 24), but zeroing six successes credited from accumulated site state removes that exception (Appendix A), and its triage signal is below chance.

Choosing which mode dies of label supply. The natural label, the cheapest mode that solved the task, exists only where something succeeded, so it is produced at the success rate: 15–97 labels per cell over six classes (Appendix Table 21), and under a minimum-of-ten-rows-per-class filter four of six cells admit no classifier at all (Appendix Table 22).

The scarcity is worse than a label count suggests, because the *question* is scarce too: the tasks where more than one mode succeeds, the only rows on which a per-task choice even exists, are 1.5–34.6% of a cell, and on two cells they number 3 and 4 (Appendix Table 16).

Choosing whether to spend has the labels and still buys nothing. The triage label is defined everywhere and predictable in five of the six cells (AUROC 0.651–0.717; the sixth, `red·B2`, is below chance at 0.483; Appendix Table 14), yet under fully nested cross-validation no cell's learned triage Pareto-dominates always taking the cheapest mode (Appendix Table 30).

A free signal read straight off the task text exists, and routing on it still loses. A regex over the task intent flags tasks where the screenshot is worth +22.54pp against +0.65pp elsewhere — a split that is large and significant only on the capable classifieds backbones, and null on reddit (Appendix Figure 4; per-cell intervals in Appendix Table 23). The policy built on it still loses to always-Vision, because the screenshot does not hurt on the unflagged tasks either (Appendix Table 24).

The cascade has a usable ranking and still buys nothing. Running the cheap mode first and escalating when the model's self-reported confidence is low, in the manner of Gupta et al. [2024] on the signal of Kadavath et al. [2022], beats always-rich at no operating point in any comparable cell — although against a random-escalation control that ranking is informative (Appendix Tables 25–26).

Pooling backbones restores supply but makes labels ambiguous, because different backbones disagree about the same task (Table 29; the one surviving cell is taken apart in Appendix M). Two further features fail before they start: the benchmark's own difficulty annotation adds ΔAUROC +0.0024 (Table 27), and whether the task supplies a reference image, the feature a practitioner would reach for first, looks like it should help and does not (Table 28).

The pattern is not 'no signal'. It is that the arm the router would route to is already the right arm to route everything to.

7 The gap, and what would close it

Between the rerun-corrected upper bound of §5 and the lower bound of §6 sits a gap no method in this paper crosses, and the closed routes fail for three separate reasons, supply, estimand and value, rather than one obstruction a better router might get around (Appendix Table 31; the route-by-route derivations are Appendix L). The sharpest statement of the bind: routing supervision is produced at the success rate, so routing is least learnable exactly where it would be most valuable; and the rows a router must learn from are the contested ones, which are the rows that flip between identical reruns (Appendix Table 10).

What would change the answer, in order of leverage:

- **Graded evaluators.** The benchmark emits two values (Appendix Table 41); a graded per-task score would make every episode a training signal regardless of success, attacking the supply obstruction directly.
- **Replicates as a reporting norm.** Single-run reporting remains the norm in this literature; at the flip rates we measure, a mode-to-mode difference below roughly 4pp is not resolvable by one run against another (§4). Rerun floors should be reported next to mode deltas.
- **A third workload, and cross-family coverage on the second benchmark**, so the moderator of the winner-reversal (§3) becomes identifiable.
- **Online cascade infrastructure.** Every escalation number here is an offline splice; what a real cascade does after the cheap arm has acted on a stateful site is unobserved in this project.

The supply–value coupling is falsifiable. It would be easy to read the obstructions above as separate walls: too few labels, and too few tasks where a choice even exists. They are one wall. The rows a which-mode router can be trained on exist only where something succeeded, and the rows where a per-task choice exists at all are those with more than one solver. Both are subsets of the tasks the agent can do, and across our eight cells they track the best single mode's success rate at $\rho = 0.952$, mean gap 1.65pp (Appendix Table 15). Part of that coupling is structural, both being functionals of one solve matrix. What is not structural is that the routable share grows roughly *linearly* in the per-mode rate, where mutually independent modes would make it grow near-quadratically at these success levels, since two of them would then both solve a task at roughly the square of the per-mode rate. The negative result here is therefore a statement about routing at 2–36% success rather than about routing, and it predicts its own reversal: run this measurement on an agent that solves most of these tasks and the label supply and the contested set grow together. We would rather be refuted that way than cited as evidence that representation routing does not work. Until then the number worth aiming at is the cost ceiling of §5: 9.5–30.6% in every cell, reachable with one bit per task rather than a mode identity, and reached by none of the five policies here.

8 Discussion

A builder of web agents can take four things from this without adopting our conclusion.

Measure your own cell. Which representation wins is a property of the deployment, not of the modality: it reverses between our two task sets and it is not stable under rerun noise in the low-success cells. A number read off someone else's benchmark does not transfer to your site, and the eight cells here disagree with each other more than any of them disagrees with a plausible prior.

Price a rerun before pricing a representation. The cheapest way to raise a union ceiling is to run an arm you already have a second time. Until that is measured, a gain from adding a representation cannot be attributed to the representation. Our own headline ceiling survived as a measurement but not as an attribution; the cost ceiling survived as both, because it adds no arm.

Treat a difference smaller than the rerun band as unresolved, not as small. At the flip rate we observe, a two-point difference between two modes is what a single repetition delivers on its own. Reporting it as an effect is reporting the resolution of the instrument.

State the denominator, and do not pool a bill with an energy estimate. Per-attempt and per-success cost disagree about which mode is cheapest in most of our cells; an API bill and an electricity estimate are not one quantity; and a latency figure is mostly browser and container. Each silently reorders the modes, so an efficiency claim without its estimand is ambiguous rather than weak: the trouble is not that the effect is small, it is that one cannot tell what was measured.

None of this settles whether representation routing can be made to work. It bounds the question: what a perfect choice would buy, what the available supervision delivers, and how far apart the two sit on the workloads the field measures.

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A Threats to validity

What the benchmark hands us. The evaluator emits two values, so there is no graded target and every routing negative inherits that (Table 41); rule-based trajectory scoring is known to disagree with human judgement often enough to matter [Lù et al., 2025]. WebArena has since been re-audited task by task [El hattami et al., 2025] and the field's reported progress reassessed on live sites [Xue et al., 2025], but the VisualWebArena audit below is our own.

What the harness adds. An arm's measured ceiling is partly this harness. Vision is on the coordinate path by construction, and that path succeeds far less often than the element-id one (Table 38). Reddit episodes leave the benchmark for the public internet on 1.05–2.13% of steps against 0.16% on classifieds, and reddit's container is slower than classifieds' before any agent behaviour enters (Table 39); Table 40 corrects a page-change false positive.

What we did ourselves. Six successes were credited by accumulated site state, and zeroing them flips one verdict (Appendix Table 42; the full earned/leaked audit, now including WebArena at 0 leaked of 37 audited episodes, a lower bound, is Table 43).

Provenance. The threat we can report on rather than only declare is our own. An audit of the analysis code found five conclusions hardcoded in the producers that generate the tables stating the same quantity: four were stale denominators, and the fifth was wrong on the fact, asserting that Vision is the cheapest mode in every cell when on `WA·B0` it is DOM. Each was true when written and never re-derived as the data grew. The fix is mechanical rather than editorial: every number in this paper and its appendix is now read from a product JSON when the table is rendered, so a stale claim cannot survive a regeneration. We report the count as a lower bound, since the sweep that found them matched a single textual shape.

B Limitations

- **No third workload.** The planned shopping workload never launched, so every claim in this paper rests on classifieds and reddit task sets alone.
- **No cross-family control on WebArena.** Both WA cells are Qwen; the cross-family control exists only on VWA. This data cannot hold "across benchmarks" and "across families" simultaneously.
- **The two benchmarks share one application.** WA reddit *is* the `vwa-reddit` container: same image, same port, same account. On the reddit axis "two benchmarks" means one application with two task sets.
- **The cascade outcome is an offline splice.** No run observes what a real cascade does after the cheap arm has already acted on a stateful site.
- **Energy is uncalibrated** (psutil at ~66W on a device rated several times that) and the local per-token cost constant was derived for a different accelerator.
- **The rerun band is measured in two cells and imported into six.** Three arms of `cls·B0` and five modes of `wa_red·B1` carry a replicate; no VWA-reddit cell and no B2 cell carries one. Every "clears the noise floor" statement about those six assumes the floor transfers, and it is not known to be mode- or cell-invariant.
- **The WebArena leakage zero is a lower bound, not a clean bill.** No leaked success was found in 37 audited episodes; that bounds the rate rather than establishing absence.
- **Two workloads cannot identify what moderates the reversal.** Modality, task set and benchmark all change together at the VWA/WA boundary; with two levels no design separates them.
- **The B2 cells are near the floor.** They carry 1–8 successes per arm and 3–4 routable tasks, so a ratio computed on them is one or two tasks wide. They are reported for coverage and should not carry a comparison.
- **The lower bound covers five simple constructions.** Logistic triage, a regex rule, a confidence cascade, pooled tiers and which-mode selection; no LLM-based router, reinforcement learning, contextual bandit or online policy was tested. The supply obstruction is estimator-independent, but the value of richer hypothesis classes on this data is unmeasured.

Table 1: Success rate per mode. Success rate (%) per observation mode. Denominator is the canonical scored set (cls 224 / red 203) and, for WebArena, the six-mode task intersection (104). Bold = best mode in the cell. Source: `representation_class_comparison.json`.

cell	DOM	SoM	Vision	DOM+stext	DOM+sprompt	SoM-image	best
cls·B0	17.41	**27.23**	25.00	15.62	19.64	15.62	SoM
cls·B1	6.25	**14.29**	12.50	7.59	6.70	6.70	SoM
cls·B2	1.34	**2.23**	2.23	0.45	1.79	0.89	SoM
red·B0	14.29	**14.78**	7.39	13.30	12.32	10.84	SoM
red·B1	5.91	**7.39**	2.46	5.91	5.42	5.91	SoM
red·B2	**3.94**	0.99	1.97	1.97	0.00	0.49	DOM
WA·B0	26.92	22.12	19.23	**35.58**	25.96	25.00	DOM+stext
WA·B1	**16.35**	13.46	9.62	16.35	16.35	11.54	DOM

Table 2: Best arm per deployment class. Best arm within each deployment class (%). Classes: no-image = {DOM, DOM+stext, DOM+sprompt, SoM-image}, vision-only = {Vision}, hybrid = {SoM}. Grouping the four is a deployment convention, not an equivalence claim: over 26 metrics none of them is the repeated extreme (Table 5), which is a counting diagnostic rather than a separability test. Source: `representation_class_comparison.json`.

cell	no-image (4 arms)	vision-only	hybrid	sole best
cls·B0	19.64 (DOM+sprompt)	25.00	27.23	hybrid
cls·B1	7.59 (DOM+stext)	12.50	14.29	hybrid
cls·B2	1.79 (DOM+sprompt)	2.23	2.23	tie: hybrid+vision-only
red·B0	14.29 (DOM)	7.39	14.78	hybrid
red·B1	5.91 (DOM)	2.46	7.39	hybrid
red·B2	3.94 (DOM)	1.97	0.99	no-image
WA·B0	35.58 (DOM+stext)	19.23	22.12	no-image
WA·B1	16.35 (DOM)	9.62	13.46	no-image

C Multiplicity and inferential scope

Multiplicity status. The tables in this paper and its appendix are not one inferential family and are not corrected as one. They divide into three classes: confirmatory (hypotheses fixed before the data and corrected within their own family, currently the 2×2 axis-independence set of 64 conjunction hypotheses under BH/Holm [Holm, 1979] on $\max(p_1, p_2)$); descriptive (intervals, no test, no correction claimed: the full behavioural matrices, the efficiency tables, the failure-attribution counts); and selection-derived (a maximum over layers, signals, thresholds or comparators is part of the statistic, so these bound what a selection could deliver and cannot be read as effect estimates [Cawley and Talbot, 2010]; each such table says so in its caption). No family-wise statement covers the set as a whole, and none is claimed.

D Deployment classes and ablation

Table 2 is the unmatched class comparison (a maximum over four no-image arms, so quote its ordering and never its gaps); Table 4 is the class ablation, unmatched and arm-matched; Table 5 is the behavioural non-separability tally. It cannot license pooling the four text modes. The tally counts how many metrics separate a mode from the rest in at least seven of eight cells, and under a pure-noise null that bar returns a zero with probability $\approx$76%, so a mode scoring zero is the expected outcome of no effect and not evidence of sameness. What the table does establish is the positive half: Vision (9 metrics) and SoM (5) are strongly separable from the rest, and that is how the deployment-class grouping of §2 is used: as a convention that survives this evidence rather than one derived from it.

E Full behavioural matrices

The four dimensions unsummarised, for every metric in every cell and mode: Outcome, Macro (what the agent did), Micro (how often what it did failed), and Efficiency. These are the substrate the consistency counts in Table 5 are computed from, included so a reader can check a claim rather

Table 3: Deployment classes at one arm each. The same comparison at one arm per class (%). Table 2's no-image column is a maximum over four arms while the other two are single arms, so it is biased up; this panel uses the arm of each class that exists outside this study. The ordering does not move: hybrid 4, no-image 3, one tie, and vision-only is never a sole best in either version: which is the robustness statement Table 2 cannot make. The gaps do move: on `WA·B0` the no-image lead is +4.81pp here against +13.46pp there, because Table 2's figure is carried by DOM+stext. Quote this table for any class gap; Table 2 only for the ordering. Source: `representation_class_comparison.json`.

cell	no-image (DOM)	vision-only (Vision)	hybrid (SoM)	sole best	gap vs hybrid	Table 2 gap
cls·B0	17.41	25.00	27.23	hybrid	-9.82	-7.59
cls·B1	6.25	12.50	14.29	hybrid	-8.04	-6.70
cls·B2	1.34	2.23	2.23	tie: hybrid+vision-only	-0.89	-0.45
red·B0	14.29	7.39	14.78	hybrid	-0.49	-0.49
red·B1	5.91	2.46	7.39	hybrid	-1.48	-1.48
red·B2	3.94	1.97	0.99	no-image	+2.96	+2.96
WA·B0	26.92	19.23	22.12	no-image	+4.81	+13.46
WA·B1	16.35	9.62	13.46	no-image	+2.88	+2.88

Table 4: Class ablation, unmatched and arm-matched. Class ablation. Columns 2–4: oracle coverage lost when a whole class is unavailable: not arm-matched, the no-image class has four arms and the others one each, so most of the gap is arm count. Columns 5–7: the matched comparison: gain from adding ONE arm of that class to the cell's best single arm (: = that class already supplies the starting arm). The matched panel shows no systematic difference between classes. Source: `representation_class_comparison.json`.

cell	all six	−no-image	−vision-only	−hybrid	+1 no-image	+1 vision-only	+1 hybrid
cls·B0	43.30	−9.38	−4.02	−2.68	+7.14pp	+6.70pp	—
cls·B1	24.55	−5.36	−4.02	−4.46	+3.57pp	+4.91pp	—
cls·B2	7.14	−2.68	−2.23	−2.23	+1.79pp	+2.23pp	—
red·B0	26.11	−7.88	−1.97	−1.97	+4.93pp	+3.45pp	—
red·B1	11.82	−3.45	−0.99	−0.49	+1.97pp	+0.99pp	—
red·B2	7.39	−4.43	−1.48	−0.49	+0.99pp	+1.97pp	+0.49pp
WA·B0	51.92	−22.12	−3.85	−1.92	+5.77pp	+5.77pp	+4.81pp
WA·B1	30.77	−14.42	−0.96	−0.96	+4.81pp	+3.85pp	+3.85pp

Table 5: Absence of repeated extrema among image-free modes. Absence of repeated extrema: read the name literally. A mode 'reaches the bar' on a metric when it is the extreme (highest or lowest) in $\geq$7 of 8 cells (87.5%). Over 26 metrics the four image-free modes reach it on none. **Caution:** This is not a separability test. `rank_consistency()` only counts which mode attains each metric's max/min; a mode that is consistently *second* (and strongly distinguishable) never reaches the bar. Establishing non-separability would need pairwise equivalence margins with task-clustered intervals, which this does not do. **Caution:** The threshold got stricter when cells grew, and an earlier caption said otherwise: $\geq$7/8 is 87.5%, not the 83% it claimed, and the six-cell $\geq$5/6 it says it matches is 83.3%: a +4.2pp shift, chosen after the cell count changed. Carrying the literal numerator ($\geq$5/8 = 62.5%) instead would let DOM+stext clear two metrics and this negative would appear to break, so the choice is load-bearing and is disclosed rather than defended. Source: `per_mode_four_dimension_profile_with_wa.json`

mode	metrics reaching the bar
Vision	9
SoM	5
DOM	0
DOM+stext	0
DOM+sprompt	0
SoM-image	0

Table 6: Full matrix: Outcome dimension. Outcome dimension, every cell $\times$ every mode. `unique` counts tasks no other mode in that cell solved. Denominators: cls 224 / red 203 / WA 104. Source: `per_mode_four_dimension_profile_with_wa.json`.

cell	mode	SR %	solves	unique
cls·B0	DOM	17.41	39	4
cls·B0	SoM	27.23	61	6
cls·B0	Vision	25.00	56	9
cls·B0	DOM+stext	15.62	35	2
cls·B0	DOM+sprompt	19.64	44	6
cls·B0	SoM-image	15.62	35	2
cls·B1	DOM	6.25	14	1
cls·B1	SoM	14.29	32	10
cls·B1	Vision	12.50	28	9
cls·B1	DOM+stext	7.59	17	1
cls·B1	DOM+sprompt	6.70	15	2
cls·B1	SoM-image	6.70	15	3
cls·B2	DOM	1.34	3	1
cls·B2	SoM	2.23	5	5
cls·B2	Vision	2.23	5	5
cls·B2	DOM+stext	0.45	1	0
cls·B2	DOM+sprompt	1.79	4	0
cls·B2	SoM-image	0.89	2	1
red·B0	DOM	14.29	29	3
red·B0	SoM	14.78	30	4
red·B0	Vision	7.39	15	4
red·B0	DOM+stext	13.30	27	2
red·B0	DOM+sprompt	12.32	25	2
red·B0	SoM-image	10.84	22	2
red·B1	DOM	5.91	12	1
red·B1	SoM	7.39	15	1
red·B1	Vision	2.46	5	2
red·B1	DOM+stext	5.91	12	1
red·B1	DOM+sprompt	5.42	11	2
red·B1	SoM-image	5.91	12	0
red·B2	DOM	3.94	8	6
red·B2	SoM	0.99	2	1
red·B2	Vision	1.97	4	3
red·B2	DOM+stext	1.97	4	1
red·B2	DOM+sprompt	0.00	0	0
red·B2	SoM-image	0.49	1	1
WA·B0	DOM	26.92	28	2
WA·B0	SoM	22.12	23	2
WA·B0	Vision	19.23	20	4
WA·B0	DOM+stext	35.58	37	7
WA·B0	DOM+sprompt	25.96	27	2
WA·B0	SoM-image	25.00	26	1
WA·B1	DOM	16.35	17	3
WA·B1	SoM	13.46	14	1
WA·B1	Vision	9.62	10	1
WA·B1	DOM+stext	16.35	17	3
WA·B1	DOM+sprompt	16.35	17	5
WA·B1	SoM-image	11.54	12	1

than take the tally on trust. Two columns are gates rather than measurements, and the tables mark them: `loc-fallback` is near-zero for Vision because there are no element ids to fall back from, not because it fails less.

Table 7: Full matrix (Macro dimension. Macro dimension) what the agent did, per step, aggregated per episode. Fractions are over agent actions. `cap-hit` = share of episodes that exhausted the 30-step budget. Source: `per_mode_four_dimension_profile_with_wa.json`.

cell	mode	steps/ep	cap-hit	click	type	scroll	search-loop	URL-revisit
cls·B0	DOM	15.61	0.268	0.322	0.224	0.183	0.812	0.606
cls·B0	SoM	13.67	0.259	0.319	0.205	0.155	0.692	0.558
cls·B0	Vision	15.88	0.295	0.353	0.146	0.260	0.728	0.639
cls·B0	DOM+stext	15.83	0.295	0.322	0.194	0.196	0.808	0.603
cls·B0	DOM+sprompt	14.96	0.290	0.328	0.208	0.174	0.759	0.583
cls·B0	SoM-image	16.23	0.321	0.312	0.201	0.208	0.790	0.600
cls·B1	DOM	21.38	0.585	0.246	0.345	0.171	0.777	0.698
cls·B1	SoM	18.01	0.491	0.368	0.211	0.137	0.643	0.655
cls·B1	Vision	20.17	0.598	0.330	0.071	0.360	0.603	0.745
cls·B1	DOM+stext	22.46	0.625	0.281	0.367	0.148	0.804	0.716
cls·B1	DOM+sprompt	21.40	0.594	0.375	0.209	0.163	0.723	0.710
cls·B1	SoM-image	21.26	0.576	0.436	0.212	0.154	0.737	0.715
cls·B2	DOM	27.38	0.817	0.520	0.142	0.031	0.647	0.823
cls·B2	SoM	24.37	0.688	0.488	0.143	0.033	0.616	0.828
cls·B2	Vision	28.25	0.888	0.336	0.128	0.323	0.710	0.896
cls·B2	DOM+stext	26.85	0.804	0.384	0.168	0.043	0.629	0.832
cls·B2	DOM+sprompt	27.84	0.830	0.514	0.078	0.048	0.629	0.857
cls·B2	SoM-image	28.38	0.853	0.523	0.082	0.033	0.683	0.858
red·B0	DOM	20.18	0.498	0.455	0.148	0.166	0.473	0.721
red·B0	SoM	20.08	0.512	0.474	0.123	0.099	0.363	0.731
red·B0	Vision	23.55	0.672	0.339	0.079	0.343	0.239	0.817
red·B0	DOM+stext	23.22	0.647	0.442	0.153	0.117	0.388	0.775
red·B0	DOM+sprompt	19.87	0.463	0.475	0.142	0.126	0.428	0.704
red·B0	SoM-image	22.90	0.592	0.453	0.137	0.124	0.338	0.775
red·B1	DOM	23.44	0.680	0.471	0.238	0.066	0.596	0.754
red·B1	SoM	22.38	0.670	0.574	0.129	0.095	0.409	0.747
red·B1	Vision	23.24	0.685	0.434	0.029	0.270	0.148	0.812
red·B1	DOM+stext	25.56	0.783	0.476	0.246	0.067	0.606	0.793
red·B1	DOM+sprompt	23.72	0.695	0.545	0.149	0.052	0.567	0.757
red·B1	SoM-image	25.17	0.773	0.580	0.166	0.056	0.557	0.778
red·B2	DOM	28.39	0.897	0.724	0.108	0.042	0.227	0.870
red·B2	SoM	26.34	0.783	0.702	0.103	0.043	0.148	0.857
red·B2	Vision	26.91	0.828	0.349	0.081	0.344	0.138	0.911
red·B2	DOM+stext	27.27	0.818	0.660	0.099	0.038	0.266	0.885
red·B2	DOM+sprompt	27.68	0.842	0.703	0.081	0.052	0.271	0.873
red·B2	SoM-image	27.87	0.882	0.708	0.094	0.048	0.217	0.882
WA·B0	DOM	16.88	0.298	0.544	0.208	0.072	0.308	0.709
WA·B0	SoM	17.45	0.365	0.465	0.278	0.029	0.327	0.716
WA·B0	Vision	22.38	0.567	0.409	0.200	0.241	0.240	0.783
WA·B0	DOM+stext	19.77	0.471	0.478	0.323	0.045	0.288	0.741
WA·B0	DOM+sprompt	17.08	0.317	0.538	0.198	0.068	0.260	0.714
WA·B0	SoM-image	18.97	0.385	0.502	0.284	0.056	0.221	0.729
WA·B1	DOM	22.64	0.625	0.459	0.319	0.023	0.644	0.746
WA·B1	SoM	23.91	0.702	0.499	0.218	0.035	0.452	0.793
WA·B1	Vision	23.12	0.606	0.450	0.097	0.249	0.260	0.807
WA·B1	DOM+stext	23.33	0.615	0.495	0.316	0.028	0.683	0.733
WA·B1	DOM+sprompt	24.50	0.702	0.551	0.231	0.033	0.519	0.776
WA·B1	SoM-image	23.79	0.683	0.569	0.249	0.017	0.587	0.772

F Replicates and run-to-run noise

G Efficiency and its estimands

Two tables carry the efficiency point. Table 12 decomposes a latency figure: the model call is 22–67% of a step, the rest is browser and container, and stripping the environment share changes which mode is fastest in 4 of 8 cells. Table 13 switches the denominator from per-attempt to per-success and the cheapest mode changes in 4 of the 6 cells with enough successes to divide by, with every pairwise interval overlapping. Both support the same statement: an efficiency claim without its estimand is ambiguous, and the estimand is usually left implicit.

H Routing, triage, and cascade

Table 14 is the triage-label predictability that §6 builds on. Table 15 asks whether the two obstructions that section reports, too few labels and too few contested tasks, are two walls or one: both sets are subsets of the tasks something solves, and across the eight cells they track the best single mode's

Table 8: Full matrix (Micro dimension. Micro dimension) per-step execution quality. `act-fail|click` and `act-fail|type` are conditional on the action type, so they are not comparable to the unconditional `act-fail` column. **Caution:** They are also not pure conditionals. An episode containing no click at all has no click-failure rate to report, and the producer stores `0.0` there (an undefined rate encoded as a perfect one) which is then averaged over every task. The zero-denominator share is large: 25–35% of episodes never type, and a few percent never click. The product now carries `*_fail_rate_complete_case` (averaged only over episodes where the action occurred) and `*_fail_rate_denom_zero_frac` beside these columns; the complete-case values run higher, and any statement about which mode fails most on clicks or types should be read from those, not from this column. **Caution:** `loc-fallback` is near-zero for Vision by construction (no element ids to fall back from), not as a finding. Source: `per_mode_four_dimension_profile_with_wa.json`.

cell	mode	parse-fail	act-fail	act-fail	click	act-fail	type	no-op	scroll-inert	no-op	success	vis-gap
cls·B0	DOM	0.0007	0.133	0.163	0.035	0.241	0.096	0.108	0.046	0.078	0.324	0.737
cls·B0	SoM	0.0021	0.082	0.102	0.007	0.216	0.018	0.134	0.051	0.027	0.300	0.741
cls·B0	Vision	0.0000	0.150	0.153	0.005	0.263	0.072	0.113	0.058	0.002	0.396	0.705
cls·B0	DOM+stext	0.0000	0.101	0.100	0.022	0.209	0.071	0.108	0.042	0.046	0.310	0.705
cls·B0	DOM+sprompt	0.0009	0.133	0.159	0.026	0.246	0.078	0.114	0.049	0.079	0.322	0.719
cls·B0	SoM-image	0.0005	0.112	0.130	0.016	0.221	0.078	0.109	0.042	0.037	0.332	0.688
cls·B1	DOM	0.0031	0.285	0.155	0.153	0.359	0.219	0.074	0.011	0.175	0.443	0.411
cls·B1	SoM	0.0016	0.249	0.203	0.068	0.353	0.149	0.104	0.013	0.118	0.390	0.509
cls·B1	Vision	0.0000	0.454	0.345	0.031	0.548	0.278	0.094	0.018	0.011	0.530	0.402
cls·B1	DOM+stext	0.0004	0.259	0.182	0.140	0.317	0.159	0.058	0.010	0.202	0.461	0.384
cls·B1	DOM+sprompt	0.0044	0.309	0.272	0.081	0.380	0.178	0.070	0.012	0.174	0.409	0.406
cls·B1	SoM-image	0.0010	0.322	0.274	0.136	0.381	0.169	0.059	0.015	0.203	0.461	0.424
cls·B2	DOM	0.0259	0.509	0.555	0.207	0.526	0.099	0.018	0.030	0.339	0.630	0.125
cls·B2	SoM	0.0876	0.559	0.475	0.110	0.584	0.067	0.025	0.026	0.327	0.627	0.152
cls·B2	Vision	0.0067	0.670	0.773	0.355	0.685	0.443	0.015	0.017	0.124	0.518	0.103
cls·B2	DOM+stext	0.0761	0.441	0.438	0.227	0.453	0.130	0.012	0.029	0.292	0.602	0.062
cls·B2	DOM+sprompt	0.0518	0.495	0.569	0.167	0.504	0.137	0.009	0.028	0.312	0.633	0.071
cls·B2	SoM-image	0.0522	0.425	0.507	0.109	0.433	0.099	0.008	0.027	0.273	0.649	0.054
red·B0	DOM	0.0002	0.222	0.194	0.051	0.294	0.161	0.072	0.075	0.119	0.428	0.507
red·B0	SoM	0.0008	0.252	0.127	0.022	0.328	0.085	0.077	0.092	0.068	0.491	0.488
red·B0	Vision	0.0000	0.382	0.193	0.018	0.430	0.268	0.049	0.064	0.006	0.556	0.328
red·B0	DOM+stext	0.0005	0.292	0.126	0.062	0.339	0.146	0.047	0.087	0.064	0.518	0.358
red·B0	DOM+sprompt	0.0026	0.194	0.146	0.046	0.270	0.149	0.076	0.093	0.083	0.421	0.537
red·B0	SoM-image	0.0027	0.285	0.118	0.034	0.335	0.160	0.050	0.097	0.070	0.502	0.408
red·B1	DOM	0.0021	0.227	0.210	0.049	0.274	0.090	0.047	0.050	0.143	0.465	0.315
red·B1	SoM	0.0063	0.297	0.228	0.062	0.357	0.111	0.061	0.061	0.176	0.552	0.320
red·B1	Vision	0.0141	0.532	0.334	0.010	0.582	0.302	0.050	0.022	0.004	0.632	0.291
red·B1	DOM+stext	0.0018	0.283	0.195	0.107	0.319	0.156	0.036	0.042	0.164	0.511	0.217
red·B1	DOM+sprompt	0.0010	0.295	0.278	0.056	0.340	0.114	0.045	0.055	0.201	0.494	0.305
red·B1	SoM-image	0.0019	0.332	0.297	0.077	0.368	0.146	0.036	0.059	0.208	0.544	0.222
red·B2	DOM	0.0224	0.492	0.502	0.111	0.501	0.156	0.009	0.044	0.443	0.770	0.074
red·B2	SoM	0.0431	0.384	0.344	0.090	0.403	0.053	0.019	0.052	0.305	0.760	0.163
red·B2	Vision	0.0163	0.640	0.613	0.276	0.669	0.547	0.029	0.005	0.078	0.597	0.153
red·B2	DOM+stext	0.0786	0.285	0.263	0.081	0.292	0.094	0.007	0.041	0.202	0.728	0.059
red·B2	DOM+sprompt	0.0660	0.604	0.621	0.126	0.609	0.115	0.005	0.042	0.503	0.724	0.044
red·B2	SoM-image	0.0501	0.316	0.297	0.123	0.325	0.137	0.009	0.034	0.240	0.725	0.049
WA·B0	DOM	0.0008	0.277	0.286	0.083	0.363	0.046	0.085	0.226	0.165	0.455	0.702
WA·B0	SoM	0.0063	0.249	0.139	0.079	0.345	0.026	0.096	0.219	0.045	0.433	0.625
WA·B0	Vision	0.0000	0.245	0.178	0.042	0.295	0.165	0.050	0.152	0.007	0.478	0.433
WA·B0	DOM+stext	0.0010	0.347	0.292	0.200	0.421	0.057	0.074	0.247	0.099	0.497	0.538
WA·B0	DOM+sprompt	0.0018	0.295	0.278	0.070	0.379	0.045	0.085	0.223	0.157	0.469	0.692
WA·B0	SoM-image	0.0033	0.302	0.268	0.170	0.377	0.040	0.075	0.268	0.092	0.500	0.625
WA·B1	DOM	0.0227	0.244	0.201	0.038	0.289	0.026	0.045	0.135	0.139	0.512	0.327
WA·B1	SoM	0.0052	0.363	0.214	0.084	0.415	0.024	0.052	0.166	0.149	0.573	0.279
WA·B1	Vision	0.0072	0.416	0.326	0.023	0.456	0.215	0.041	0.102	0.002	0.571	0.365
WA·B1	DOM+stext	0.0102	0.206	0.197	0.076	0.247	0.045	0.041	0.113	0.105	0.503	0.385
WA·B1	DOM+sprompt	0.0068	0.343	0.302	0.070	0.379	0.066	0.036	0.098	0.197	0.522	0.298
WA·B1	SoM-image	0.0144	0.343	0.315	0.076	0.382	0.036	0.039	0.099	0.180	0.562	0.288

Table 9: Full matrix (Efficiency dimension. Efficiency dimension. Cost is comparable within a cell only) B0 bills a proxy API, B1/B2 are electricity-derived from a per-token constant calibrated for a different accelerator, so absolute dollars for B1/B2 are uncalibrated and only `cost rel DOM` is safe across backbones. `latency canon` removes retry, busy-wait and recovered-screenshot time; it differs from raw only on the API-served arm. Source: `per_mode_four_dimension_profile_with_wa.json`.

cell	mode	cost/ep	cost rel DOM	latency s	latency canon s	tokens/ep
cls·B0	DOM	0.06962	1.000	115.0	114.1	63045
cls·B0	SoM	0.07236	1.039	106.7	106.0	67000
cls·B0	Vision	0.06481	0.931	126.3	125.2	58302
cls·B0	DOM+stext	0.06919	0.994	123.8	120.4	62154
cls·B0	DOM+sprompt	0.06853	0.984	109.4	107.8	62555
cls·B0	SoM-image	0.07206	1.035	121.1	117.9	65334
cls·B1	DOM	0.05951	1.000	308.9	308.9	61904
cls·B1	SoM	0.06028	1.013	262.0	262.0	62953
cls·B1	Vision	0.04316	0.725	269.8	269.8	44524
cls·B1	DOM+stext	0.05879	0.988	313.5	313.5	61074
cls·B1	DOM+sprompt	0.06304	1.059	301.4	301.4	65628
cls·B1	SoM-image	0.05970	1.003	311.7	311.7	61985
cls·B2	DOM	0.07676	1.000	402.3	402.3	79653
cls·B2	SoM	0.09075	1.182	374.3	374.3	95081
cls·B2	Vision	0.07065	0.920	417.8	417.8	73126
cls·B2	DOM+stext	0.07320	0.954	399.4	399.4	75946
cls·B2	DOM+sprompt	0.08453	1.101	396.4	396.4	87948
cls·B2	SoM-image	0.08456	1.102	411.0	411.0	87931
red·B0	DOM	0.10147	1.000	572.0	552.5	93303
red·B0	SoM	0.11045	1.089	461.4	451.6	103031
red·B0	Vision	0.09807	0.966	449.8	418.5	88872
red·B0	DOM+stext	0.10577	1.042	631.4	562.1	96214
red·B0	DOM+sprompt	0.10163	1.002	498.4	447.7	93817
red·B0	SoM-image	0.10814	1.066	562.0	532.0	99005
red·B1	DOM	0.07330	1.000	602.6	602.6	76462
red·B1	SoM	0.08000	1.091	609.1	609.1	83517
red·B1	Vision	0.05240	0.715	456.6	456.6	53994
red·B1	DOM+stext	0.06948	0.948	598.6	598.6	72222
red·B1	DOM+sprompt	0.07656	1.044	614.0	614.0	79896
red·B1	SoM-image	0.07480	1.020	616.6	616.6	77722
red·B2	DOM	0.09479	1.000	669.9	669.9	99015
red·B2	SoM	0.11160	1.177	623.0	623.0	117379
red·B2	Vision	0.06833	0.721	550.0	550.0	70826
red·B2	DOM+stext	0.08852	0.934	677.6	677.6	92346
red·B2	DOM+sprompt	0.09940	1.049	599.3	599.3	104021
red·B2	SoM-image	0.09451	0.997	640.0	640.0	98699
WA·B0	DOM	0.07531	1.000	284.1	282.7	68616
WA·B0	SoM	0.09110	1.210	272.1	271.7	84854
WA·B0	Vision	0.08640	1.147	337.9	337.5	77786
WA·B0	DOM+stext	0.08478	1.126	330.3	328.4	76591
WA·B0	DOM+sprompt	0.07747	1.029	262.4	260.9	71176
WA·B0	SoM-image	0.08498	1.128	324.0	320.9	77689
WA·B1	DOM	0.06579	1.000	485.2	485.2	68491
WA·B1	SoM	0.07944	1.208	494.8	494.8	82940
WA·B1	Vision	0.04468	0.679	490.7	490.7	45878
WA·B1	DOM+stext	0.06151	0.935	509.9	509.9	63727
WA·B1	DOM+sprompt	0.07386	1.123	506.3	506.3	76839
WA·B1	SoM-image	0.06659	1.012	510.6	510.6	68819

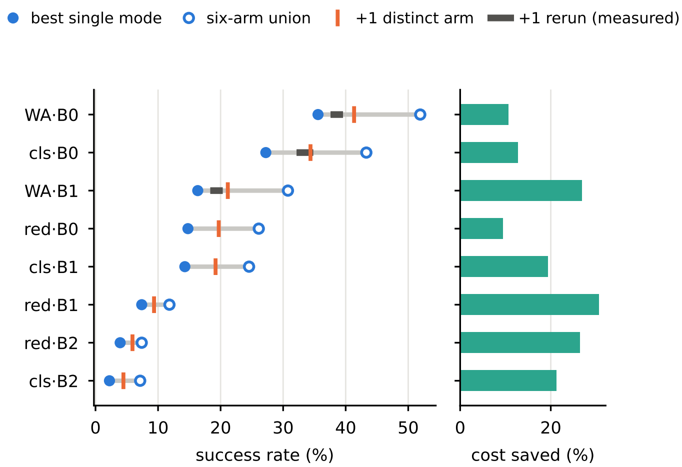

Figure 3: The two ceilings, per cell (full table: Appendix Table 16). Left: the six-run union ceiling against the best single mode; the tick is what one genuinely different arm adds, and the dark segment is what one rerun of an arm already in hand adds, where measured. Most of the apparent headroom is not attributable to representation diversity. Right: the cost ceiling at matched success, the bound that survives, because it adds no arm.

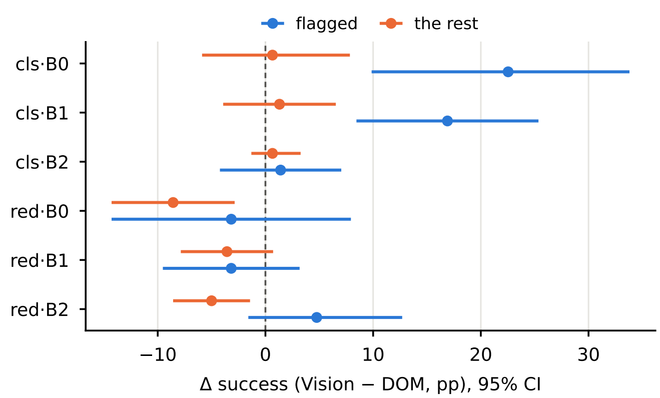

Figure 4: The zero-token partition (Appendix Table 23): the screenshot's worth (Vision − DOM) on regex-flagged tasks versus the rest. It is large and significant only on the capable classifieds backbones, and null on reddit. The predicate flags 71 of 224 classifieds tasks and 63 of 203 reddit tasks.

success rate at $\rho = 0.952$. Part of that association is structural rather than empirical, and the caption separates the two; what it licenses is the framing of §7, not a new effect.

Table 24 turns the ex-ante partition (Table 23) into a policy. Table 25's cascade beats always-rich at no operating point in any comparable cell, and Table 26 is the table that says what that means: against the *same budget spent at random* the confidence ranking wins nearly everywhere, so the signal is informative and still not enough. Table 27 adds the benchmark's own difficulty annotation for a mean ΔAUROC of +0.0024. Table 28 shows the feature a practitioner would reach for first looks like it should help and does not. Table 29 pools the backbones and routes by cost tier instead of by task.

The bound the rest of this section does not reach is Table 16 in §5. The oracle stated against the best single mode, and its triage/route decomposition, are below, followed by the rerun comparator (Table 19), the per-cell fusion-premium intervals behind Figure 5 (Table 20), the which-mode label supply (Table 21), and the per-cell partition intervals behind Figure 4 (Table 23).

Table 10: Per-task label instability. Per-task label instability on `cls_B0` (n=224): 49 tasks change outcome between two runs of the same condition. The rows a which-mode router could learn from are exactly the contested ones, and they carry almost all of the instability. **Caution: this table is computed on 2 replicated arms of one cell (B0.cls.dom, B0.cls.vision), rerun once.** The project's whole replicate inventory is three arms of that cell (the SoM pair landed after this product was generated and is not folded in here) plus five modes of `wa_red_B1` on a registered ten-task draw; no VWA-reddit cell and no B2 cell carries one at all, so every stability figure elsewhere is imported rather than measured. The headline enrichment has two defensible definitions and neither may be quoted alone: 17.4x defined over all 6 arms is correct for the claim (a router chooses among 6) but the flips are produced by rerunning 2 of them, so the same arms decide both membership and outcome; rebuilding the difficulty proxy from the other 4 breaks that circle and gives 3.95x. Source: `label_instability.json`.

stratum	tasks	share of cell	flipped	flip rate	share of all flips	enrichment vs complement
which-mode label rows (any mode solved)	97	43.3%	47	48.45%	95.9%	**16.47x**
. . . of those, the arms DISAGREE (the choice matters)	88	39.3%	45	51.14%	91.8%	**17.39x**
three-way channel decision is contested	74	33.0%	36	48.65%	73.5%	**16.54x**
exactly one mode solved it (label unambiguous)	29	12.9%	15	51.72%	30.6%	**17.59x**
complement: no mode solved, or all did	136	60.7%	4	2.94%	8.2%	**1.00x**
whole cell	224	100.0%	49	21.88%	100.0%	**7.44x**

I Failure attribution and mechanisms

Table 34 locates *when* two modes part on the tasks whose outcome they disagree about: at the first or second step, essentially never late, which is not the shape accumulated drift has. It is markedly weaker on WebArena, so it is reported per site rather than pooled. Table 33 is the paired cut: on tasks only one channel solved, how did the other fail? It counts rule hits, so Table 32 asks the same question without the rule vocabulary, using six probes read straight off the step records, and finds the losing channel failing *more blandly* there than elsewhere, which is what closes the objection that the residual is an artefact of a VWA-shaped ruleset. Table 35 decomposes the DOM→SoM-image transition into a text axis and a prompt axis, and Table 36 asks whether that text axis moves per-step decision quality more than it moves macro action frequencies. Table 37 counts hallucinated element references, which are inapplicable to Vision by construction, and the table marks them. **Caution:** a per-rule frequency is a distribution of symptoms, not of causes. The two largest rows in most cells are risk markers that causal verification did not confirm as death causes.

Each block of Table 33 states the largest enrichment among the rules it does *not* list. Read each block against its own baseline column, never across blocks: the text channel is four arms and the image channel two, so the two blocks' task counts are not comparable to each other.

J Audits and provenance

The dispatch-path audit (Table 38), the off-site navigation and container-latency audit (Table 39), the page-change detector correction (Table 40), the evaluator granularity check (Table 41), and the full earned-versus-leaked audit including the first WebArena pass (Table 43).

Table 11: Behavioural metrics against run-to-run noise. Behavioural metrics against run-to-run movement, `B0 x classifieds`, three replicated arms (dom, vision, som). `rerun band` is the largest |metric(run A) - metric(run B)| over those arms; `cross-mode spread` is max-min over the six modes. 22 of 25 metrics exceed the band, several by 5-22x. The exceptions are `mean_latency_canonical_s` (0.84x), `mean_latency_s` (0.87x), `parse_fail_rate` (1.00x): i.e. both latency metrics, which `latency_decomposition` reaches independently by decomposing the step into model and container. Every other efficiency and behavioural claim in this paper is judged against a rerun band; these 26 metrics were not, until this table. One cell, one rerun per arm: a point estimate, not a threshold. Source: `replicate_metric_noise.json`.

dimension	metric	cross-mode spread	rerun band	ratio	> a rerun?
Outcome	`sr_pct`	11.607	2.232	5.20x	**yes**
Outcome	`n_success`	26.000	5.000	5.20x	**yes**
Outcome	`n_unique_solves`	—	—	—	*cross-mode by construction*
Macro	`n_steps`	2.567	0.469	5.48x	**yes**
Macro	`cap_hit_rate`	0.062	0.045	1.40x	**yes**
Macro	`click_frac`	0.041	0.014	2.87x	**yes**
Macro	`type_frac`	0.078	0.019	4.21x	**yes**
Macro	`scroll_frac`	0.106	0.006	16.85x	**yes**
Macro	`search_loop_rate`	0.121	0.018	6.75x	**yes**
Macro	`url_revisit_rate`	0.081	0.022	3.66x	**yes**
Micro	`parse_fail_rate`	0.002	0.002	1.00x	no
Micro	`action_fail_rate`	0.068	0.014	4.87x	**yes**
Micro	`click_fail_rate`	0.063	0.025	2.48x	**yes**
Micro	`type_fail_rate`	0.030	0.006	5.28x	**yes**
Micro	`no_change_rate`	0.053	0.009	5.82x	**yes**
Micro	`scroll_inert_rate`	0.078	0.017	4.68x	**yes**
Micro	`noop_inert_rate`	0.026	0.007	3.92x	**yes**
Micro	`visibility_gap_rate`	0.016	0.011	1.56x	**yes**
Micro	`locator_fallback_rate`	0.076	0.006	11.81x	**yes**
Micro	`action_repeat_frac`	0.096	0.004	21.88x	**yes**
Micro	`finish_rate`	0.054	0.036	1.50x	**yes**
Efficiency	`mean_cost_usd`	0.008	0.002	3.24x	**yes**
Efficiency	`cost_rel_dom`	0.108	0.026	4.22x	**yes**
Efficiency	`mean_latency_s`	19.562	22.488	0.87x	no
Efficiency	`mean_latency_canonical_s`	19.204	22.846	0.84x	no
Efficiency	`mean_tokens`	8697.978	2186.054	3.98x	**yes**

K Label definitions, nesting, and identifiability

Every number in these tables is quoted in the body section that references it. They are placed here because the body has an eight-page limit and these tables document the results rather than making the argument.

K.1 Observation modes

The grid §2 refers to. The last column is the modality split Appendix L.4 uses. It is not a cost split, for the reason §2 gives: the image-bearing tier holds both the dearest mode and the cheapest.

Table 12: What a latency number contains. What a latency number contains. Every latency figure elsewhere in this paper is the whole step; `backend_infer` isolates the model call and was read by no analysis script until 2026-08-03. The model is 22–67% of the measured time; the rest is the browser and the container, which `offsite_navigation_audit` measures at 1.69× between the two sites. Removing it changes which mode is fastest in 4 of 8 cells, and not at random: 4 of 5 reddit-family cells flip against 0 of 3 classifieds cells: the flips land where the container is slowest. A sentence naming the fastest mode is therefore partly a sentence about this deployment. What survives estimand choice is only that the two orderings disagree. Source: `latency_decomposition.json`.

cell	mean step (ms)	model call (ms)	model share	fastest by total	fastest by model only	same?
B0·classifieds	7,622	2,140	**28.1%**	DOM+sprompt	DOM+sprompt	yes
B0·reddit	24,601	5,603	**22.8%**	Vision	SoM-image	**no**
B0·wa_reddit	16,112	3,551	**22.0%**	Vision	DOM+sprompt	**no**
B1·classifieds	14,180	8,213	**57.9%**	Vision	Vision	yes
B1·reddit	24,376	7,916	**32.5%**	Vision	DOM+stext	**no**
B1·wa_reddit	21,221	7,747	**36.5%**	DOM+sprompt	Vision	**no**
B2·classifieds	14,739	9,908	**67.2%**	DOM+sprompt	DOM+sprompt	yes
B2·reddit	22,863	9,131	**39.9%**	Vision	Vision	yes

Table 13: Per-attempt versus per-success. Per-attempt versus per-success denominators. `content? = no` marks cells whose best mode has fewer than 10 successes: their ratios are directions at best and both B2 cells are in that state. Among the 6 cells with content the cheapest mode changes under the success denominator in 4 and the fastest in 3. **Caution:** Every pairwise CI overlaps, so this supports the methodological point that the denominator must be declared, not any ranking. Source: `outcome_efficiency.json`.

cell	content?	cheapest/attempt	cheapest/success	fastest/attempt	fastest/success	max solves
cls·B0	yes	Vision	Vision	SoM	SoM	61
red·B0	yes	Vision	DOM	Vision	SoM	30
cls·B1	yes	Vision	Vision	SoM	SoM	32
red·B1	yes	Vision	SoM	Vision	SoM	15
cls·B2	**no**	Vision	Vision	SoM	SoM	5
red·B2	**no**	Vision	DOM	Vision	DOM	8
WA·B1	yes	Vision	DOM+stext	DOM	DOM	17
WA·B0	yes	DOM	DOM+stext	DOM+sprompt	DOM+stext	37

Table 14: Predictability of the triage label. AUROC is cross-validated; the comparison column is the best single covariate used alone, so the margin isolates what the multivariate model adds. The reddit · B2 row is below chance and is the subject of §6.

cell	tasks	solvable	AUROC (logistic)	best single covariate	margin
classifieds · B0	224	43.3%	0.676	0.607	+0.069
reddit · B0	203	26.1%	0.666	0.612	+0.054
classifieds · B1	224	24.6%	0.717	0.627	+0.090
reddit · B1	203	11.8%	0.685	0.637	+0.048
classifieds · B2	224	7.1%	0.651	0.655	−0.005
reddit · B2	203	7.4%	0.483	0.711	−0.228

Table 15: Label supply and routing value are one quantity. Label supply and routing value are the same set. The rows a which-mode router can be trained on exist only where something succeeded; the rows where a per-task choice exists at all are those with more than one solver. Both are subsets of the solvable set, and across all 8 cells they track the best single mode's success rate with Spearman rho = 0.952 (exact permutation p = 0.0011), mean |best SR - routable share| = 1.65pp. So the two obstructions the lower bound reports are not independent walls but one wall, whose height is set by how much the agent can do. **Caution: part of this association is structural, not empirical**: both columns are functionals of one solve matrix. The non-structural content is the near-linearity: under mode independence the routable share would grow near-quadratically in the per-mode rate at these success levels, and it does not, which says task difficulty dominates mode-task matching. **Caution: the ratio is not constant**: 0.53-0.71 in the 6 cells above the floor, but half that (`red_B2`, `cls_B2`) on 3 and 4 routable tasks, where it is one task wide. **Caution: the cells are not independent**: they share three sites and three backbones, so the permutation null asks whether the pairing is arbitrary, not whether eight systems were sampled. Source: `supply_value_coupling.json`.

cell	n	best single mode	best SR	routable set (>1 solver)	solvable (any mode)	routable / solvable
`wa_red_B0`	104	DOM+stext	35.58%	34.6% (36)	51.9%	0.67
`cls_B0`	224	SoM	27.23%	30.4% (68)	43.3%	0.70
`wa_red_B1`	104	DOM	16.35%	17.3% (18)	30.8%	0.56
`red_B0`	203	SoM	14.78%	17.7% (36)	26.1%	0.68
`cls_B1`	224	SoM	14.29%	12.9% (29)	24.6%	0.53
`red_B1`	203	SoM	7.39%	8.4% (17)	11.8%	0.71
`red_B2`	203	DOM	3.94%	1.5% (3)	7.4%	0.20
`cls_B2`	224	SoM	2.23%	1.8% (4)	7.1%	0.25

Table 16: What a perfect per-task choice could buy. Two ceilings, and only one of them survives its own control. *Ceiling: any mode solves it* is what a perfect per-task choice could reach; it runs 7.1-51.9% against a best single mode of 2.2-35.6%. **Caution:** That column is a six-arm union against a one-arm baseline. The arm-matched comparison is the last two columns: adding the single best distinct arm buys +1.97 to +7.14pp, and rerunning an arm already in hand buys a draw in the same range wherever a replicate exists – so the headroom cannot be attributed to representation diversity rather than to resampling. The union of *five* reruns has never been measured, so the split at higher arm counts is unknown, not estimated. *Same tasks, lower cost* keeps the best mode everywhere and sends only the tasks no mode solves to the cheapest one: success is unchanged by construction and cost falls 9.5-30.6% in 8 of 8 cells. That ceiling is immune to the arm-count objection because it adds no arms. Why both are hard to reach: no mode solves 48.1-92.9% of tasks, and the set where a per-task choice even exists is 1.5-34.6% of the cell. Leaked successes are kept here, as in every other table; 6 scored successes on reddit were credited without the episode ever visiting the forum the evaluator reads, and zeroing them (denominator unchanged) is reported as a sensitivity analysis rather than folded in: the detection criterion is a lower bound, so it cannot license a corrected point estimate. Both sets of figures are in the product. Source: `routing_ceiling.json`

cell	n	best single mode	ceiling: any mode solves	headroom	same tasks, lower cost	+1 arm	rerun once
WA·B0	104	DOM+stext 35.58%	**51.92%**	+16.35pp	**-10.7%**	+5.77	2.00-4.00
cls·B0	224	SoM 27.23%	**43.30%**	+16.07pp	**-12.8%**	+7.14	4.91-7.59
WA·B1	104	DOM 16.35%	**30.77%**	+14.42pp	**-26.8%**	+4.81	2.00-4.00
red·B0	203	SoM 14.78%	**26.11%**	+11.33pp	**-9.5%**	+4.93	–
cls·B1	224	SoM 14.29%	**24.55%**	+10.27pp	**-19.4%**	+4.91	–
red·B1	203	SoM 7.39%	**11.82%**	+4.43pp	**-30.6%**	+1.97	–
red·B2	203	DOM 3.94%	**7.39%**	+3.45pp	**-26.4%**	+1.97	–
cls·B2	224	SoM 2.23%	**7.14%**	+4.91pp	**-21.3%**	+2.23	–

Table 17: The routing ceiling, over classifieds (cls) and reddit (red). The last three columns give success rate (%) and mean per-episode billed cost (USD). The oracle picks the cheapest mode that solved each task, so its success rate equals the solvable column by construction; "cheapest" is the lowest-mean-cost mode applied everywhere. Cost is never compared across backbones.

cell	n	solvable	best single	cheapest	oracle
cls · B0	224	43.3%	27.23 / 0.07236	25.00 / 0.06481	43.30 / 0.05777
red · B0	203	26.1%	14.78 / 0.11045	7.39 / 0.09807	26.11 / 0.09534
cls · B1	224	24.6%	14.29 / 0.06028	12.50 / 0.04316	24.55 / 0.04171
red · B1	203	11.8%	7.39 / 0.08000	2.46 / 0.05240	11.82 / 0.05178
cls · B2	224	7.1%	2.23 / 0.09075	2.23 / 0.07065	7.14 / 0.06953
red · B2	203	7.4%	3.94 / 0.09479	1.97 / 0.06833	7.39 / 0.06958

Table 18: The ceiling decomposed. Both columns are deltas against the best single mode. The triage half is accuracy-neutral by construction and carries 9.5–30.6% of cost saving; the route half carries 3.45–16.35pp of success rate and much less cost. The two WebArena rows were added 2026-08-04 and are the favourable end of both halves: `WA·B0` carries the largest route-half headroom (+16.35pp) and the largest route-half cost saving (−16.2%) anywhere in the study, so the six-cell version understated the ceiling on the cells where routing has most to work with. The two halves require different labels, and the rest of the paper fails on one obstruction per half.

cell	triage half: ΔSR / Δcost	route half: ΔSR / Δcost
classifieds · B0	±0.00pp / −12.8%	+16.07pp / −7.4%
reddit · B0	±0.00pp / −9.5%	+11.33pp / −4.2%
classifieds · B1	±0.00pp / −19.4%	+10.27pp / −11.4%
reddit · B1	±0.00pp / −30.6%	+4.43pp / −4.7%
classifieds · B2	±0.00pp / −21.3%	+4.91pp / −2.1%
reddit · B2	±0.00pp / −26.4%	+3.45pp / −0.2%
WA reddit · B0	±0.00pp / −10.7%	+16.35pp / −16.2%
WA reddit · B1	±0.00pp / −26.8%	+14.42pp / −6.7%

Table 19: New representation versus a rerun. Is a new representation worth more than a rerun? Both middle columns are the same functional at the same arm count (`|{added} \ {baseline}| / n`) so they are directly comparable; only the *source* of the extra arm differs. The band is 3 rerun pairs on one cell (`B0 × classifieds`, n=224), one each for DOM, SoM, Vision: so the rows without a band have no comparator at all, and the band itself is 3 draws rather than a bound. Since the SoM replicate landed 2026-08-03 the band is no longer extrapolated onto an unreplicated arm: both the fused mode this table's best-single column keeps selecting and the arm the comparison adds now carry their own measured floor, and adding the third pair left the band unmoved. Source: `noise_floor_inventory.json`

cell	best single	+1 distinct arm	+1 rerun (measured floor)	verdict
cls·B0	SoM at 27.23	+7.14 (DOM)	4.91 – 7.59pp	**inside the rerun band**
cls·B1	SoM at 14.29	+4.91 (Vision)	—	no floor on this cell
cls·B2	SoM at 2.23	+2.23 (Vision)	—	no floor on this cell
red·B0	SoM at 14.78	+4.93 (DOM)	—	no floor on this cell
red·B1	SoM at 7.39	+1.97 (DOM+sprompt)	—	no floor on this cell
red·B2	DOM at 3.94	+1.97 (Vision)	—	no floor on this cell
WA·B0	DOM+stext at 35.58	+5.77 (DOM)	—	no floor on this cell
WA·B1	DOM at 16.35	+4.81 (DOM+stext)	0.00 – 10.00pp *(pooled would read 2.00–4.00)*	**inside the rerun band**

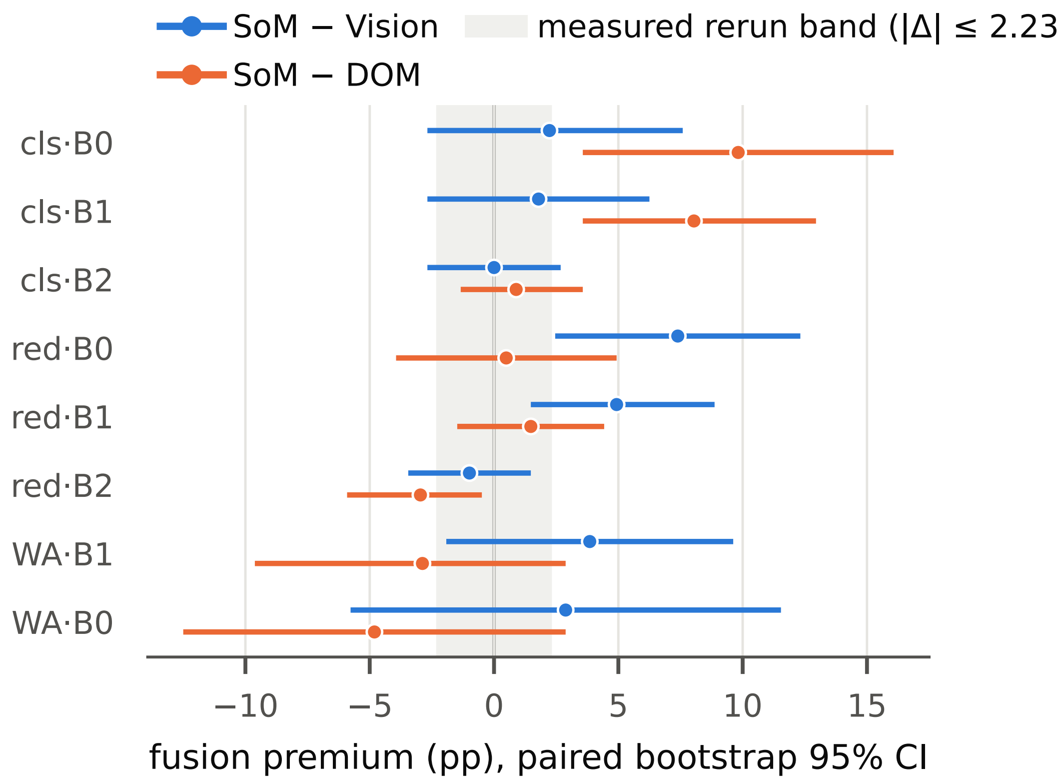

Figure 5: Fusion premium per cell (Appendix Table 20), paired bootstrap 95% CIs. The shaded region is the rerun threshold of §4: 3.8–4.2pp, the level a mean difference must reach before a single repetition of one arm would be unlikely to produce it. Against the workload-matched channel no cell clears it (the largest such premium is +2.23pp); SoM's decisive wins are all against the channel mismatched to that workload.

Table 20: Fusion premium against the rerun band. Fusion premium (pp). Comparators are fixed a priori, not per-cell maxima. Paired bootstrap over tasks, 10,000 resamples. Read against the measured rerun band 0.89–2.23pp, not against zero: a premium must beat what repetition delivers for the same money. No cell clears the band; `cls_B0`'s +2.23 *equals* its upper edge (both are 5/224). Source: `fusion_premium.json`

cell	n	SoM − Vision	95% CI	SoM − DOM	95% CI
cls·B0	224	+2.23	[-2.68, +7.59]	+9.82	[+3.57, +16.07]
cls·B1	224	+1.79	[-2.68, +6.25]	+8.04	[+3.57, +12.95]
cls·B2	224	+0.00	[-2.68, +2.68]	+0.89	[-1.34, +3.57]
red·B0	203	+7.39	[+2.46, +12.32]	+0.49	[-3.94, +4.93]
red·B1	203	+4.93	[+1.48, +8.87]	+1.48	[-1.48, +4.43]
red·B2	203	-0.99	[-3.45, +1.48]	-2.96	[-5.91, -0.49]
WA·B1	104	+3.85	[-1.92, +9.62]	-2.88	[-9.62, +2.88]
WA·B0	104	+2.88	[-5.77, +11.54]	-4.81	[-12.50, +2.88]

Table 21: Which-mode label supply. A label exists only where some mode solved the task, so the labelled column equals the number of solved tasks. Six classes have to be discriminated from between 15 and 97 examples.

cell	scored tasks	labelled	solvable	classes present
classifieds · B0	224	97	43.3%	6/6
reddit · B0	203	53	26.1%	6/6
classifieds · B1	224	55	24.6%	6/6
reddit · B1	203	24	11.8%	6/6
classifieds · B2	224	16	7.1%	5/6
reddit · B2	203	15	7.4%	5/6

Table 22: Classes surviving a minimum of ten training rows per class in a five-fold split. Fewer than two surviving classes leaves nothing to discriminate between, which is the state of four of the six cells.

cell	labels	classes	classes surviving the filter	trainable
classifieds · B0	97	6	3 (DOM, DOM+sprompt, SoM)	yes
reddit · B0	53	6	1 (DOM)	no
classifieds · B1	55	6	2 (DOM, SoM)	yes
reddit · B1	24	6	0	no
classifieds · B2	16	5	0	no
reddit · B2	15	5	0	no

Table 23: Ex-ante visual-intent partition. Ex-ante partition. The predicate is a regex over the task intent plus 'carries no reference image': both read the task config, so it costs no tokens and needs no episode. On classifieds the screenshot is worth an order of magnitude more on the flagged tasks than on the rest, and the flagged/rest split is significant/not respectively on the two capable backbones. **Caution:** WebArena is omitted: the predicate fires on only 5 of 104 tasks there and none is solved by any mode, so the cells are degenerate rather than null. Source: `visual_intent_routing.json`.

cell	flagged	arm	Δ vs DOM on flagged	95% CI	Δ on the rest	95% CI
cls·B0	71	vision	+22.54	[+9.86, +33.80]	+0.65	[-5.88, +7.84]
cls·B0	71	som	+19.72	[+7.04, +32.39]	+5.23	[-1.31, +12.42]
cls·B1	71	vision	+16.90	[+8.45, +25.35]	+1.31	[-3.92, +6.54]
cls·B1	71	som	+12.68	[+5.63, +21.13]	+5.88	[+0.00, +11.76]
cls·B2	71	vision	+1.41	[-4.23, +7.04]	+0.65	[-1.31, +3.27]
cls·B2	71	som	+0.00	[-5.63, +5.63]	+1.31	[-1.31, +3.92]
red·B0	63	vision	-3.17	[-14.29, +7.94]	-8.57	[-14.29, -2.86]
red·B0	63	som	+0.00	[-9.52, +9.52]	+0.71	[-4.29, +5.71]
red·B1	63	vision	-3.17	[-9.52, +3.17]	-3.57	[-7.86, +0.71]
red·B1	63	som	+1.59	[-3.17, +7.94]	+1.43	[-2.14, +5.00]
red·B2	63	vision	+4.76	[-1.59, +12.70]	-5.00	[-8.57, -1.43]
red·B2	63	som	-1.59	[-4.76, +0.00]	-3.57	[-7.14, +0.00]

K.2 Where the which-mode label disagrees with measured cost

Per-cell detail for the 12.5–54.6% range quoted in §6, and for the claim that the exact-tie case never occurs.

K.3 Effect of nesting the operating-point selection

Per-cell detail for the -0.99pp to +1.34pp range quoted in §6, and for the observation that nesting moves the result in both directions.

K.4 Identifiability of pooled which-mode labels

Per-site detail for the conflict rates and modal-agreement figures quoted in Appendix L.3.

Table 24: Routing policies on the 3-axis frontier. Is routing worth it? `rule` policies send the ex-ante-flagged tasks to one arm and the rest to another; the partition is a regex over the task intent, so nothing is learned and there is no in-sample optimism. 'On frontier' means nothing dominates it, not that it is preferable: on `cls·B0` all three rule policies sit between always-SoM and always-Vision, worse on every axis than one or the other. Cost/latency are per-attempt cell means, within-cell comparable only. Source: `rule_routing_pareto.json`.

cell	policy	SR	cost	latency	on frontier
cls·B0	always-SoM	27.23	0.07236	106.0s	yes
cls·B0	always-Vision	25.00	0.06481	125.2s	yes
cls·B0	rule: flag→Vision else DOM	24.55	0.06809	117.6s	yes
cls·B0	rule: flag→SoM else DOM+stext	24.11	0.07019	115.8s	yes
cls·B0	rule: flag→SoM else DOM	23.66	0.07049	111.5s	yes
cls·B0	always-DOM	17.41	0.06962	114.1s	no
cls·B1	always-SoM	14.29	0.06028	262.0s	yes
cls·B1	always-Vision	12.50	0.04316	269.8s	yes
cls·B1	rule: flag→Vision else DOM	11.61	0.05433	296.5s	no
cls·B1	rule: flag→SoM else DOM+stext	11.16	0.05926	297.2s	no
cls·B1	rule: flag→SoM else DOM	10.27	0.05976	294.0s	no
cls·B1	always-DOM	6.25	0.05951	308.9s	no
cls·B2	always-SoM	2.23	0.09075	374.3s	yes
cls·B2	always-Vision	2.23	0.07065	417.8s	yes
cls·B2	rule: flag→Vision else DOM	1.79	0.07483	407.2s	yes
cls·B2	always-DOM	1.34	0.07676	402.3s	yes
cls·B2	rule: flag→SoM else DOM	1.34	0.08120	393.4s	no
cls·B2	rule: flag→SoM else DOM+stext	1.34	0.07876	391.5s	yes
red·B0	always-SoM	14.78	0.11045	451.6s	yes
red·B0	always-DOM	14.29	0.10147	552.5s	yes
red·B0	rule: flag→SoM else DOM	14.29	0.10426	521.2s	yes
red·B0	rule: flag→Vision else DOM	13.30	0.10041	510.9s	yes
red·B0	rule: flag→SoM else DOM+stext	13.30	0.10722	527.8s	no
red·B0	always-Vision	7.39	0.09807	418.5s	yes
red·B1	always-SoM	7.39	0.08000	609.1s	yes
red·B1	rule: flag→SoM else DOM+stext	6.90	0.07275	601.8s	yes
red·B1	rule: flag→SoM else DOM	6.40	0.07538	604.6s	no
red·B1	always-DOM	5.91	0.07330	602.6s	no
red·B1	rule: flag→Vision else DOM	4.93	0.06682	557.3s	yes
red·B1	always-Vision	2.46	0.05240	456.6s	yes
red·B2	rule: flag→Vision else DOM	5.42	0.08658	632.7s	yes
red·B2	always-DOM	3.94	0.09479	669.9s	no
red·B2	rule: flag→SoM else DOM	3.45	0.10001	655.4s	no
red·B2	always-Vision	1.97	0.06833	550.0s	yes
red·B2	rule: flag→SoM else DOM+stext	1.48	0.09568	660.7s	no
red·B2	always-SoM	0.99	0.11160	623.0s	no

We call it agreement and not a Bayes ceiling because it is a resubstitution estimate: it scores the same rows it took the modal label from, so a task labelled by only one backbone is correct by construction. That describes 50 of 168 classifieds rows (29.8%) and 34 of 92 reddit rows (37.0%); restricted to tasks two or more backbones label, agreement falls to 70.3% and 74.1%. An out-of-sample bound would need leave-one-backbone-out prediction or a shrinkage estimator, and would be lower still. Every number here is therefore an optimistic bound on what a pooled classifier could reach, which is the direction the argument needs.

The last column groups by the exact feature vector instead of by task. Rows of one task are not always identical, because three of the five numeric features are read from that backbone's own step-0 observation, so they differ somewhere on 31.5% of shared classifieds tasks and 80.0% of shared reddit tasks. We report but do not use that grouping: it leaves 74 of 117 classifieds groups and 69 of 78 reddit groups with a single member, covering 44% and 75% of rows, so it is even more inflated than the headline, and a router serving one backbone could not recover backbone identity from that jitter anyway.

Table 25: Confidence-triggered cascade. Confidence-triggered cascade, vision $\rightarrow$ som. The escalation decision sees only the cheap run's own episode: no outcome, no rich-run information. In 0 of the 6 comparable cells does any operating point beat always-rich on success rate; the cells marked *rich is worse* are excluded because the cascade's premise fails there, and that exclusion rule (`cheap_sr >= rich_sr`) is outcome-dependent. **Caution:** The two counts are split for a reason. A point also 'wins' by matching always-rich's SR at lower cost, and every such point here is a tie artefact: on `WA·B1` all of them sit at exactly the rich arm's SR while 44-60 of 104 episodes are tied at the cutoff, so the ranking falls through to task id and the reachable SR spans 8.65-14.42. Counting the two together is what let this table print a WA win for months while the prose said the WA exception was withdrawn: the rows disagreed with the sentence and nobody could see it, because the WA rows were dropped before render until 2026-08-03. **Caution:** Every number is an offline splice: an escalated task takes its outcome from a standalone rich run, whereas a real cascade would start the rich episode after the cheap one had already acted on a stateful site. That sequential outcome is unobserved in this project. Source: `confidence_cascade_with_wa.json`

cell	n	cheap SR	rich SR	always-rich cost	oracle SR	comparable?	beats always-rich — strictly / SR-tied	signals dropped
cls·B0	224	25.00	27.23	1.116×	33.93	yes	0 / 0	2
cls·B1	224	12.50	14.29	1.397×	19.20	yes	0 / 0	0
cls·B2	224	2.23	2.23	1.284×	4.46	— *(rich is worse)*	**11** / 22	0
red·B0	203	7.39	14.78	1.126×	18.23	yes	0 / 0	0
red·B1	203	2.46	7.39	1.527×	8.37	yes	0 / 0	2
red·B2	203	1.97	0.99	1.633×	2.96	— *(rich is worse)*	**41** / 5	1
WA·B0	104	19.23	22.12	1.054×	29.81	yes	0 / 0	0
WA·B1	104	9.62	13.46	1.778×	16.35	yes	0 / 3	1

Table 26: Cascade signal against a random-escalation control. Is the cascade's signal doing anything? Table 25 answers a deployment question: no operating point beats always-rich on success rate: which on its own cannot distinguish a signal that carries nothing from one that carries something insufficient. This is the comparator that separates them: the same escalation budget spent at random. **Caution:** Read the median column, not the best column. The best margin is positive in 24 of 24 (cell $\times$ fraction) combinations, but each of those is a maximum over that cell's 8–10 candidate signals against a constant (the random comparator does not depend on which signal is used) so positivity there is partly arithmetic. Over all 222 (signal $\times$ cell $\times$ fraction) points, 64.4% are positive with a median of +0.327pp: that is the unselected statement, and it is the one to quote. On cls·B2, red·B0, WA·B1 the median is negative at one or more fractions while the best is positive: there the apparent win is entirely selection. The right-hand column is how much of the gap to a per-task oracle the best signal recovers, and inherits the same caveat. The offline-splice caveat on Table 25 applies unchanged. Source: `confidence_cascade_with_wa.json`.

cell	signals	**best** margin (10/20/30%)	**median over all signals** (10/20/30%)	signals >0	oracle headroom captured (10/20/30%)
cls·B0	8	+1.12 / +1.78 / +2.01	+1.12 / +1.11 / +0.67	19/24	15% / 25% / 30%
cls·B1	10	+1.16 / +1.43 / +2.14	+0.27 / +0.53 / +0.81	19/30	20% / 27% / 40%
cls·B2	10	+0.45 / +0.89 / +0.45	+0.00 / +0.00 / +0.00	11/30	20% / 40% / 20%
red·B0	10	+1.74 / +1.46 / +0.74	+0.01 / +0.48 / -0.25	15/30	23% / 27% / 27%
red·B1	8	+1.49 / +1.96 / +1.48	+0.50 / +1.22 / +1.23	23/24	33% / 50% / 50%
red·B2	9	+0.10 / +0.20 / +0.30	+0.10 / +0.20 / +0.30	20/27	0% / 0% / 0%
WA·B0	10	+3.57 / +3.26 / +3.95	+1.16 / +1.82 / +1.54	24/30	36% / 36% / 45%
WA·B1	9	+2.51 / +2.11 / +2.70	-0.37 / -0.78 / -0.18	12/27	43% / 43% / 57%

Table 27: Triage learnability and the visual-difficulty feature. Can the triage label be predicted, and does the benchmark's own visual-difficulty annotation help? Task-held-out 5-fold CV, L2 logistic regression, seed 42. Mean ΔAUROC = +0.0024 over 6 cells, improving 1: inside fold-split noise. `solvable` is the positive count: the label exists for every task, unlike the which-mode label. **Caution:** Six cells, not eight, and the reason is the finding. `visual_difficulty` is a VisualWebArena task-config annotation; WebArena's 106 reddit configs carry none of `visual_difficulty`, `reasoning_difficulty`, `overall_difficulty` or `image`: the field simply is not there, so the contrast cannot be computed rather than computing to zero. That absence is what makes WA the clean test elsewhere: where a router can read the benchmark's own difficulty annotation it looks learnable, and WA is the setting where it cannot. Source: `visual_difficulty_router.json`.

cell	n	solvable	AUROC without	AUROC with visual_ difficulty	Δ	visual_ difficulty alone
cls·B0	224	97	0.726	0.726	-0.000	0.534
cls·B1	224	55	0.732	0.726	-0.006	0.553
cls·B2	224	16	0.642	0.630	-0.013	0.570
red·B0	203	53	0.780	0.776	-0.005	0.630
red·B1	203	24	0.864	0.863	-0.002	0.620
red·B2	203	15	0.615	0.655	+0.040	0.718

Table 28: The intuitive routing feature. The intuitive routing feature. A task shipping a reference image ought to route to a mode that can see images: the table shows which mode is actually best on each side of that split. **Caution:** Reference images are delivered in every mode, so this feature does not separate what it appears to. WebArena cannot arbitrate: its 106 reddit configs carry no `image` field at all (nor any of the three difficulty annotations) so both of this table's stratifiers are undefined there. Six cells is the whole population for this question, not a coverage gap. Source: `routing_feature_diagnostics.json`.

cell	tasks with ref image	without	best mode WITH image	SR	best mode WITHOUT	SR
cls·B0	65	159	SoM	40.00	Vision	22.64
cls·B1	65	159	DOM+stext	16.92	SoM	13.84
cls·B2	65	159	SoM	4.62	DOM	1.89
red·B0	79	124	SoM	26.58	DOM	7.26
red·B1	79	124	SoM	13.92	SoM	3.23
red·B2	79	124	DOM	8.86	Vision	3.23

K.5 The screenshot-modality tier

Per-site detail for Appendix L.4, on the same pooled labelled rows and the same grouping as Appendix K.4. The agreement column is the exception: it is defined only on tasks labelled by two or more backbones, since agreement needs two labels to compare.

L Derivations for the four relabelling routes

§6 and §7 state the outcome of each route. This appendix gives the derivation.

L.1 Continuous labels

The cleanest fix would be to regress on a graded quality signal rather than classify a discrete winner: partial credit turns every episode into a training example regardless of whether it succeeded. VisualWebArena does not provide one. Across the 7,686 scored episodes of our 36 landed conditions the evaluator emits exactly two values, 0.0 and 1.0, in a 7,041 / 645 split, as do the 36 protocol-excluded episodes we do not score. This is a property of the benchmark's evaluation design rather than of our pipeline, and it forecloses the route entirely.

Table 29: Pooled tier router. Pooling backbones restores label supply and does not buy a router. Labels are scarce per cell, so this arm pools backbones that share a task and routes by the coarsest label available (which of two cost tiers to spend) giving classifieds 152/reddit 77/wa_reddit 86 labelled rows against the 15–97 a single cell affords. H-pool NOT supported: the same-family $\times$ cost-tier router dominates always-cheapest in 0/6 cells and is dominated by the fixed-mode menu in every cell. The most favourable corner (agreeing backbones, coarse label, highest ceiling) does not change the negative result. Δ SR is not the story: the cost column is. Where the router does raise success it also raises cost by 2.7–55.7%, so it is buying success at a price rather than routing to a better arm, and it strictly dominates always-cheapest in 0 of 14 arms. **Caution: non-dominance is not dominance.** The `non-dominated` column is an admissibility criterion (a policy buying success for more money passes it) and it must never be quoted as a win rate. **Caution: ‡ marks rows that are identical by construction, not by result**: WebArena has no cross-family backbone, so its all-three pool and its same-family pool are the same pool, reported twice to keep the layout parallel. **Caution:** the pooled task overlap is 50 tasks on classifieds and 20 on reddit, so every per-cell contrast is underpowered; costs are never compared across backbones (an API bill against an electricity estimate); and this probe is post-hoc and exploratory: it is not the preregistered gate and must not be cited as one. Source: `router_pooled_tier_learnability.json`.

pool	cell	router SR	cheapest SR	Δ SR	Δ cost	non-dom.
same-family	`cls·B0`	25.00%	25.00%	+0.00pp	+13.9%	46.8%
same-family	`cls·B1`	12.05%	12.50%	-0.45pp	+33.0%	33.6%
all-three	`cls·B0`	24.11%	25.00%	-0.89pp	+14.6%	34.5%
all-three	`cls·B1`	11.16%	12.50%	-1.34pp	+36.6%	21.2%
all-three	`cls·B2`	1.79%	2.23%	-0.45pp	+4.8%	18.0%
same-family	`red·B0`	14.29%	7.39%	+6.90pp	+10.2%	99.2%
same-family	`red·B1`	4.43%	2.46%	+1.97pp	+38.2%	86.8%
all-three	`red·B0`	13.30%	7.39%	+5.91pp	+2.7%	98.2%
all-three	`red·B1`	4.43%	2.46%	+1.97pp	+35.8%	86.8%
all-three	`red·B2`	1.97%	1.97%	+0.00pp	+7.4%	42.5%
same-family	`wa_red·B0`	30.77%	26.92%	+3.85pp	+6.3%	83.3%
same-family	`wa_red·B1`	15.38%	9.62%	+5.77pp	+55.7%	94.1%
all-three ‡	`wa_red·B0`	30.77%	26.92%	+3.85pp	+6.3%	83.3%
all-three ‡	`wa_red·B1`	15.38%	9.62%	+5.77pp	+55.7%	94.1%

Table 30: The nested triage policy against the always-cheapest fixed policy. Pareto dominance requires no worse on both axes; no cell achieves it. The parenthetical gives the router's deltas relative to the fixed policy.

cell	nested SR / cost	always-cheapest SR / cost	Pareto-dominates?
classifieds · B0	26.79% / 0.07197	25.00% / 0.06481	no (+1.79pp SR, +11.0% cost)
reddit · B0	12.81% / 0.09836	7.39% / 0.09807	no (+5.42pp SR, +0.3% cost)
classifieds · B1	14.29% / 0.05757	12.50% / 0.04316	no (+1.79pp SR, +33.4% cost)
reddit · B1	5.91% / 0.06970	2.46% / 0.05240	no (+3.45pp SR, +33.0% cost)
classifieds · B2	1.34% / 0.07247	2.23% / 0.07065	no (−0.89pp SR, +2.6% cost)
reddit · B2	3.94% / 0.06964	1.97% / 0.06833	no (+1.97pp SR, +1.9% cost)

Table 31: The five supervision targets against the two requirements a trainable router needs. Each closed route is closed for a different reason, which is what makes the negative result closed rather than provisional. Supply and identifiability are judged on that target's own denominator, which differs between the last two rows.

route	supply	identifiability	outcome
which-mode, per cell	fails (15–97 labels, 4/6 cells untrainable)	fine	closed by §6
continuous target	would fix	fine	closed: benchmark score is binary
pooled which-mode	fixed (260)	fails (56–57% conflict; in-sample modal agreement 79–84%)	wrong estimand
triage (binary)	fine (203–224, every task)	fine	learnable, and beaten by a fixed policy (§6)
screenshot tier	fine, but only on solved tasks	fine (68–88% cross-backbone agreement, over solved tasks)	agreement measured; no tier classifier trained

Table 32: Vocabulary-free probes on the text-wins residual. Is the text-wins residual real, or an artefact of the rule vocabulary? Tables 24–25 count rule hits, and the ruleset was discovered on VisualWebArena: so an absent signature there could be a property of the vocabulary rather than of the world. Each probe here is computed from raw step fields and never from a rule hit, over 142 disagreement episodes against 2304 baseline failures. The largest enrichment is 1.15× and 5 of 6 sit *below* 1: on the tasks the text channel uniquely solves, the image channel fails more blandly than it fails elsewhere: it did not arrive, rather than breaking somewhere nameable. **Caution:** Six candidates chosen by us, so this cannot show that no mechanism exists; what it closes is the specific objection that the residual is an artefact of a VWA-shaped vocabulary. Source: `conditional_failure_attribution.json`

candidate mechanism	on the disagreement set	that channel's baseline	enrichment
finished in five steps or fewer	0.162	0.141	**1.15**×
any parse failure	0.092	0.109	**0.84**×
never searched	0.387	0.468	**0.83**×
ran out of budget (>=30 steps)	0.535	0.658	**0.81**×
page unchanged on over half the steps	0.331	0.431	**0.77**×
action failure on over half the steps	0.310	0.416	**0.74**×

L.2 Coarser classes

§6's obstruction is that four of six cells have fewer than ten labelled rows in more than one class. Merging classes does not add rows. The binary collapse of the screenshot tier (Appendix L.4) does help, but for a different reason, having to do with agreement across backbones rather than with class count.

L.3 Pooling across backbones

Six cells at 15–97 labels become 260 pooled examples and every class clears the minimum-count filter, so supply is solved. Identifiability is not. The features carry no model identity, so two backbones facing the same task on the same site produce near-identical feature vectors, and where they are identical and the oracle labels differ, a classifier is being asked to emit two different answers for one input. The figure reported in Appendix K.4 is the accuracy of emitting the modal label per group, scored on the rows the label came from. It is an in-sample bound on what a pooled classifier could reach, not a Bayes ceiling, and Appendix K.4 gives the resubstitution caveat.

They are only near-identical, not identical, because `dom_complexity`, `text_length` and `tokens_input_text` are read from the backbone's own step-0 observation rather than from the task config. On 31.5% of shared classifieds tasks and 80.0% of shared reddit tasks the rows therefore differ somewhere. Grouping by the exact vector rather than by task raises the figure (Appendix K.4, last

Table 33: Failure modes are asymmetric across channels. The two channels do not fail the same way. On tasks only one channel solved, how did the other fail? Pooled over 8 cells at ruleset v11. Enrichment = hit rate on the disagreement set over that channel's hit rate across all its failures, so about 1x means it failed there the way it fails everywhere. Rules with fewer than 8 pooled conditional hits are omitted, and the largest omitted enrichment is stated per block rather than left to the reader. Block A has named death causes (top 2.31x, 404 losing-channel episodes against 5388 of that channel's failures overall). Block B does not: its top row rests on 8 hits, exactly the reporting floor, and all 8 of them fall in the 2 WebArena cells, none in the other 6; every other rule in that block sits at or below the everywhere-baseline. On the tasks the text channel uniquely solves, the image channel did not break somewhere nameable – it did not arrive. **Caution:** TEXT is four arms and IMAGE is two, so the two blocks' task counts are not comparable to each other; read each block against its own baseline column, never across blocks. **Caution:** a per-rule frequency is a distribution of symptoms, not of causes – the largest rows in most cells are risk markers, not death causes, and only rules whose docstrings record a causal check are verified as such. Source: `conditional_failure_attribution.json`.

	rule	how it failed	on disagreement	baseline	enrichment	hits
A. image wins		*how the text channel failed*	(101 tasks)			
	`P27`	gives up when not found	3.2%	1.4%	**2.31x**	13
	`P17`	click-back oscillation	15.1%	6.7%	**2.25x**	61
	`P16`	visual task, DOM cannot see	6.2%	2.8%	**2.24x**	25
	`P43`	page visual, no screenshot	48.5%	29.3%	**1.65x**	196
		15 further rules			all <= 1.55x	
B. text wins		*how the image channel failed*	(109 tasks)			
	`P49`	submit-page anchor misclick	3.7%	1.0%	**3.61x**	8
	`P17`	click-back oscillation	4.6%	3.9%	**1.17x**	10
	`P12`	never paginates	13.8%	14.8%	**0.93x**	30
	`P31`	budget exhausted, unfinished	49.5%	54.2%	**0.91x**	108
		6 further rules			all <= 0.87x	

Table 34: Where two representations part company. Where two representations part company. Restricted to the tasks whose outcome the two modes disagree on (exactly one of them succeeds) so these are the trajectories that produced the difference, not all trajectories. `first divergent step` is the first step at which the two URL signatures differ. Drift would look different: two runs that separate through accumulated perturbation share a prefix and part late, whereas on VisualWebArena 72–100% of these tasks have already diverged by step 3 and at most 3.0% diverge at step 10 or later. **Caution: this is markedly weaker on WebArena** (50–89% early, up to 22.2% late), which is why the table is per site and not pooled. **Caution:** each row rests on 13–46 tasks, and element ids are deliberately not used as the anchor: they are neither step-invariant nor mode-invariant, so URL signatures are the only stable comparison. Source: `mechanism_per_task.json` E2.

site	contrast	tasks	median first divergent step	diverged by step 3	diverged at step 10+
classifieds	DOM vs SoM-image	22	1	95.5%	0.0%
classifieds	DOM vs DOM+ sprompt	33	1	84.8%	3.0%
classifieds	DOM vs DOM+ stext	22	1	95.5%	0.0%
classifieds	SoM-image vs SoM	46	1	93.5%	0.0%
classifieds	DOM+sprompt vs SoM-image	23	1	91.3%	0.0%
classifieds	DOM+stext vs SoM-image	18	2	72.2%	0.0%
reddit	DOM vs SoM-image	19	0	100.0%	0.0%
reddit	DOM vs DOM+ sprompt	18	1	88.9%	0.0%
reddit	DOM vs DOM+ stext	16	0	93.8%	0.0%
reddit	SoM-image vs SoM	20	0	94.7%	0.0%
reddit	DOM+sprompt vs SoM-image	17	0	100.0%	0.0%
reddit	DOM+stext vs SoM-image	19	0	100.0%	0.0%
wa_reddit	DOM vs SoM-image	16	1	75.0%	6.2%
wa_reddit	DOM vs DOM+ sprompt	19	1	89.5%	5.3%
wa_reddit	DOM vs DOM+ stext	21	1	66.7%	9.5%
wa_reddit	SoM-image vs SoM	19	1	73.7%	0.0%
wa_reddit	DOM+sprompt vs SoM-image	13	0.5	83.3%	8.3%
wa_reddit	DOM+stext vs SoM-image	19	4	50.0%	22.2%

Table 35: 2x2 axis decomposition. 2×2 axis decomposition. Effect sizes are Cohen's h (binary metrics) or d_z (paired continuous), signed right-minus-left. The compound DOM→SoM-image transition decomposes into a text-payload axis and a prompt-style axis; the image axis is SoM-image→SoM. **Caution:** On mean differences the two decomposition routes agreeing is an algebraic identity, so a zero residual is arithmetic and not evidence about an interaction. `B2 × wa_reddit` is absent because B2 never ran WebArena. Source: `axis_effect_size_with_wa.json`.

cell	metric	text axis	prompt axis	image axis	DOM→SoM-image
cls·B0	search_loop	-0.011	-0.045	-0.225	-0.056
cls·B0	type_frac	-0.169	+0.049	+0.030	-0.140
cls·B0	scroll_frac	+0.072	+0.063	-0.253	+0.134
cls·B0	selfcorr_count	+0.088	-0.115	-0.091	-0.020
cls·B0	click_frac	-0.001	-0.059	+0.035	-0.056
cls·B0	finish_rate	-0.070	-0.039	+0.119	-0.109
cls·B0	n_steps	+0.023	+0.039	-0.246	+0.066
cls·B0	action_repeat_frac	-0.051	+0.086	-0.117	+0.030
cls·B1	search_loop	+0.066	-0.160	-0.203	-0.094
cls·B1	type_frac	+0.088	-0.546	-0.005	-0.452
cls·B1	scroll_frac	-0.097	+0.025	-0.062	-0.065
cls·B1	selfcorr_count	-0.015	+0.007	+0.149	-0.007
cls·B1	click_frac	+0.139	+0.526	-0.169	+0.614
cls·B1	finish_rate	-0.055	+0.082	+0.170	+0.027
cls·B1	n_steps	+0.101	-0.108	-0.253	-0.011
cls·B1	action_repeat_frac	+0.053	+0.002	-0.174	+0.048
cls·B2	search_loop	-0.037	+0.113	-0.141	+0.076
cls·B2	type_frac	+0.116	-0.410	+0.297	-0.290
cls·B2	scroll_frac	+0.154	-0.112	+0.003	+0.026
cls·B2	selfcorr_count	+0.100	-0.111	-0.105	-0.024
cls·B2	click_frac	-0.476	+0.485	-0.097	+0.011
cls·B2	finish_rate	-0.217	-0.038	+0.333	-0.256
cls·B2	n_steps	-0.052	+0.160	-0.363	+0.115
cls·B2	action_repeat_frac	-0.093	+0.159	-0.067	+0.064
red·B0	search_loop	-0.170	-0.104	+0.052	-0.275
red·B0	type_frac	+0.029	-0.098	-0.084	-0.067
red·B0	scroll_frac	-0.206	+0.041	-0.148	-0.182
red·B0	selfcorr_count	+0.279	-0.065	-0.258	+0.223
red·B0	click_frac	-0.057	+0.053	+0.081	-0.009
red·B0	finish_rate	-0.299	+0.102	+0.160	-0.200
red·B0	n_steps	+0.276	-0.032	-0.252	+0.245
red·B0	action_repeat_frac	+0.226	-0.056	-0.033	+0.202
red·B1	search_loop	+0.020	-0.100	-0.297	-0.080
red·B1	type_frac	+0.032	-0.329	-0.159	-0.301
red·B1	scroll_frac	+0.008	-0.071	+0.157	-0.074
red·B1	selfcorr_count	-0.053	-0.075	-0.062	-0.120
red·B1	click_frac	+0.019	+0.326	-0.016	+0.370
red·B1	finish_rate	-0.224	+0.012	+0.223	-0.212
red·B1	n_steps	+0.202	-0.044	-0.231	+0.165
red·B1	action_repeat_frac	+0.133	+0.108	+0.020	+0.249
red·B2	search_loop	+0.092	-0.115	-0.179	-0.024
red·B2	type_frac	-0.045	-0.023	+0.045	-0.070
red·B2	scroll_frac	-0.037	+0.093	-0.040	+0.054
red·B2	selfcorr_count	+0.052	-0.026	+0.015	+0.019
red·B2	click_frac	-0.193	+0.143	-0.017	-0.047
red·B2	finish_rate	-0.059	-0.044	+0.382	-0.103
red·B2	n_steps	-0.126	+0.057	-0.137	-0.054
red·B2	action_repeat_frac	-0.157	-0.010	+0.121	-0.173
WA·B0	search_loop	-0.042	-0.155	+0.238	-0.197
WA·B0	type_frac	+0.476	-0.197	-0.031	+0.336
WA·B0	scroll_frac	-0.247	+0.155	-0.191	-0.135
WA·B0	selfcorr_count	+0.106	-0.053	+0.088	+0.078
WA·B0	click_frac	-0.255	+0.111	-0.139	-0.170
WA·B0	finish_rate	-0.339	+0.176	+0.000	-0.163
WA·B0	n_steps	+0.260	-0.084	-0.128	+0.200
WA·B0	action_repeat_frac	+0.157	+0.012	-0.217	+0.170
WA·B1	search_loop	+0.081	-0.200	-0.270	-0.119
WA·B1	type_frac	-0.016	-0.271	-0.109	-0.275
WA·B1	scroll_frac	+0.072	-0.133	+0.155	-0.064
WA·B1	selfcorr_count	+0.148	-0.136	+0.108	-0.006
WA·B1	click_frac	+0.153	+0.244	-0.182	+0.353
WA·B1	finish_rate	+0.121	-0.204	-0.021	-0.083
WA·B1	n_steps	+0.071	+0.058	+0.010	+0.121
WA·B1	action_repeat_frac	-0.034	+0.218	+0.029	+0.168

Table 36: Decision quality versus macro frequency. Does the text axis change per-step decisions more than it changes macro action frequencies? Ratio >1 means yes. Verdict: generalizes (site_ok = {'reddit': True, 'classifieds': True, 'wa_reddit': True}). **Caution:** `_site_ok` passes a site if any backbone clears 1.0, which is a loose bar: `WA·B0` is 0.97 and the WA site passes on B1's 2.98. Until 2026-08-03 the verdict function named only the two VWA sites literally and did not consult WA at all. Source: `axis1_microbehavior_with_wa.json`.

cell	decision effect (mean abs)	macro effect (mean abs)	ratio	>1?
cls·B0	0.1530	0.0606	**2.52**	yes
cls·B1	0.1024	0.0766	**1.34**	yes
cls·B2	0.3159	0.1556	**2.03**	yes
red·B0	0.2761	0.1926	**1.43**	yes
red·B1	0.2450	0.0863	**2.84**	yes
red·B2	0.3877	0.0952	**4.07**	yes
WA·B0	0.2284	0.2351	**0.97**	**no**
WA·B1	0.2588	0.0869	**2.98**	yes

Table 37: Hallucinated element references. Hallucinated element references, ruleset `11-intent-text-fallback` over 36 conditions. An action naming an element id that is not in the observation. **Caution:** `vision` carries no element-id list at all, so this rule is structurally inapplicable there rather than measuring zero: the same gate-versus-measurement confusion flagged for P2/P4. Source: `cross_mode_failure_signatures.json`.

cell	mode	episodes	failed	with hallucinated ref	rate of failed
classifieds·B0	DOM	224	185	15	8.1%
classifieds·B0	DOM+sprompt	224	180	11	6.1%
classifieds·B0	SoM-image	224	189	2	1.1%
classifieds·B0	DOM+stext	224	189	9	4.8%
classifieds·B0	SoM	187	129	2	1.6%
classifieds·B0	Vision	224	168	0	0.0%
reddit·B0	DOM	203	174	8	4.6%
reddit·B0	DOM+sprompt	203	178	8	4.5%
reddit·B0	SoM-image	203	181	1	0.6%
reddit·B0	DOM+stext	203	176	6	3.4%
reddit·B0	SoM	203	173	2	1.2%
reddit·B0	Vision	203	188	0	0.0%
classifieds·B1	DOM	224	210	20	9.5%
classifieds·B1	DOM+sprompt	224	209	33	15.8%
classifieds·B1	SoM-image	224	209	7	3.3%
classifieds·B1	DOM+stext	224	207	45	21.7%
classifieds·B1	SoM	224	192	0	0.0%
classifieds·B1	Vision	224	196	0	0.0%
reddit·B1	DOM	203	191	25	13.1%
reddit·B1	DOM+sprompt	203	192	33	17.2%
reddit·B1	SoM-image	203	191	3	1.6%
reddit·B1	DOM+stext	203	191	24	12.6%
reddit·B1	SoM	203	188	5	2.7%
reddit·B1	Vision	203	198	0	0.0%
classifieds·B2	DOM	224	221	82	37.1%
classifieds·B2	DOM+sprompt	224	220	114	51.8%
classifieds·B2	SoM-image	224	222	43	19.4%
classifieds·B2	DOM+stext	224	223	75	33.6%
classifieds·B2	SoM	224	219	39	17.8%
classifieds·B2	Vision	224	219	0	0.0%
reddit·B2	DOM	203	195	127	65.1%
reddit·B2	DOM+sprompt	203	203	155	76.4%
reddit·B2	SoM-image	203	202	62	30.7%
reddit·B2	DOM+stext	203	199	51	25.6%
reddit·B2	SoM	203	201	59	29.4%
reddit·B2	Vision	203	199	0	0.0%

Table 38: What actually delivered the click. How each action reached the browser. `Vision` is on the coordinate path by construction (it emits no element ids) so its action success is capped by this harness's coordinate implementation (39%) rather than by the 89% the element-id path achieves. That is not a confound to remove (it is what screenshot-only *is*), but the Vision arm measures our grounding code as much as the representation. Separately the element-id fallback share rises with backbone weakness: B0 12% · B1 35% · B2 37% on the text arms: *how often* a run falls back is a model property, the fallback's own 16% success is ours. No success rate elsewhere is adjusted by this; it bounds external validity. Source: `dispatch_path_audit.json`

delivery path	actions	action success
id locator	8,857	**88.9%**
other	1,530	**65.9%**
coord	2,138	**38.6%**
id framework	4,564	**16.1%**

Table 39: Off-site navigation and container latency. Off-site navigation. Postmill is a link aggregator, so an agent opening a trending thread can walk onto the live public internet; classifieds is self-contained. **Caution:** Off-site steps are faster, not slower (commercial CDNs beat a Postmill container sharing a host with the agent) so the distortion runs opposite to the intuition. The larger asymmetry is in the last two columns of the on-site medians: reddit's container costs ~1.69× what classifieds' does before any agent behaviour enters, which is why no between-site latency number is quotable bare. Source: `offsite_navigation_audit.json`.

			median env_step		
cell	off-site steps	off-site episodes	on-site	off-site	ratio
B0·VWA-cla	0/20646 (0.00%)	0/1344 (0.0%)	4,493 ms	—	—
B0·VWA-red	478/26425 (1.81%)	36/1230 (2.9%)	11,349 ms	10,007 ms	0.88×
B1·VWA-cla	0/27927 (0.00%)	0/1344 (0.0%)	5,750 ms	—	—
B1·VWA-red	501/29309 (1.71%)	54/1230 (4.4%)	7,848 ms	6,349 ms	0.81×
B2·VWA-cla	58/36529 (0.16%)	4/1344 (0.3%)	4,656 ms	14,884 ms	3.20×
B2·VWA-red	719/33749 (2.13%)	49/1230 (4.0%)	9,324 ms	4,850 ms	0.52×
B1·WA-red	278/14695 (1.89%)	40/624 (6.4%)	6,809 ms	6,105 ms	0.90×
B0·WA-red	123/11703 (1.05%)	21/624 (3.4%)	6,613 ms	11,405 ms	1.72×

Table 40: page_changed false positives. `page_changed` false positives. A step can register a page change that is purely cosmetic; correcting for it raises every mode's no-change rate. The Micro conclusion is unaffected (Vision remains highest in every cell either way) but a router firing on a 2-step no-change streak would trigger 5321 → 5910 times (+11.1%). Source: `page_change_corrected.json`.

	no-change rate			
mode	observed	corrected	cosmetic FP steps	steps
DOM	0.3846	0.4332	1618	33277
phantom_prompt	0.4222	0.4681	1528	33283
phantom_som	0.3683	0.4169	1688	34741
phantom_text	0.3331	0.3770	1518	34610
SoM	0.4071	0.4675	1870	30921
Vision	0.5675	0.6134	1569	34151

Table 41: Evaluator granularity. Evaluator granularity over the paper-grade set. Across 7,686 scored episodes in 36 conditions, the evaluator emits 2 distinct values (0.0, 1.0) (645 of them positive) with 0 missing and 0 non-numeric. There is no graded quality target to regress on. That is a property of the benchmark's design rather than of this pipeline, and it is a precondition of every routing negative in this document: a router's training signal can only be as fine-grained as the evaluator, so a partially-completed task is indistinguishable from one the agent never started. Source: `evaluator_score_granularity.json`.

quantity	value
conditions resolved	36
conditions unresolved	0
episodes scored	7686
episodes with no score	0
episodes with a non-numeric score	0
distinct score values	[0.0, 1.0]
count per value	{"0.0": 7041, "1.0": 645}

Table 42: Leaked-success sensitivity. Sensitivity to environmentally-credited successes. `require_reset` is a no-op on reddit, so subscriptions accumulate across a run's episodes and a later task can be scored on state an earlier one created. 6 such successes are set to 0 here: the denominator is unchanged, because an attempted-and-unaccomplished task is a 0, not a missing row. 4 of the leaks are on DOM, so removing them helps the fused arm: the direction that disfavours this project's own caution. **Caution:** The WA cells are unaudited for the same defect. Source: `leakage_sensitivity.json`.

cell	contrast	before	95% CI	after	95% CI	verdict
red·B0	SoM − Vision	+7.39	[+2.46, +12.32]	+7.88	[+2.96, +12.81]	unchanged
red·B0	SoM − DOM	+0.49	[-3.94, +4.93]	+0.99	[-3.45, +5.42]	unchanged
red·B1	SoM − Vision	+4.93	[+1.48, +8.87]	+4.43	[+0.99, +7.88]	unchanged
red·B1	SoM − DOM	+1.48	[-1.48, +4.43]	+0.99	[-1.48, +3.94]	unchanged
red·B2	SoM − Vision	-0.99	[-3.45, +1.48]	-0.99	[-3.45, +1.48]	unchanged
red·B2	SoM − DOM	-2.96	[-5.91, -0.49]	-1.48	[-3.45, +0.49]	**flips**

column), and we report but do not adopt that number: it leaves most groups with one member, and a group of one is scored perfectly whatever the labels do, so it inflates with feature sparsity rather than tracking identifiability. Every version of the number is an optimistic bound and all of them are far below what a deployable which-mode router needs, which is the only thing the argument turns on.

L.4 Screenshot tier

The tier label is derived from the same solve events as the which-mode label, by mapping each oracle mode to image-bearing (SoM, Vision) or text-only (DOM, P-text, P-prompt, P-SoM). No new episodes are involved, which is why the figure rises without a single new solve event. Part of that rise is arithmetic (two classes admit a larger modal share than six), so the agreement columns rather than the modal-agreement columns carry the claim. The tier is defined only on tasks that some mode solved, so its denominator is the solved set and not the full task universe. Appendix K.5's modal-agreement columns run over the whole pooled labelled set, as in Appendix K.4; the two agreement columns are restricted to tasks labelled by two or more backbones, that being the only set on which cross-backbone agreement is defined.

M The reddit · B2 saving in detail

§6 reports that the one cell whose cost saving survives Holm correction has an AUROC below chance. The mechanism is tail enrichment rather than a globally ordered score.

Reddit · B2 sends 192 of 203 tasks (95%) to the cheap mode with no accuracy loss, which is unsurprising in a cell where only 7.4% of tasks are solvable at all: almost nothing in that 95% was going to succeed under any mode. The policy therefore differs from the free always-cheapest policy by five percent of the allocation. The 11 tasks it holds back for the strong mode carry four

Table 43: Earned versus leaked successes. Which successes were earned. `#sidebar > section > ul` is read by 9 VWA reddit tasks; `require_reset` is a no-op on reddit so subscriptions accumulate. LEAKED = scored success by an episode that never visited the required forum. On VWA: 6 leaked, 68 earned; Table 42 recomputes every contrast with the leaked ones zeroed. WebArena audited 2026-08-03 (first time): 50 scored episodes over its 5 sidebar tasks, 0 leaked. **Caution:** That zero is a *lower bound*, not a clearance: the test asks whether the episode reached the forum, and an episode can arrive at a forum an earlier one subscribed to, read `Unsubscribe`, and finish without acting. One such case is hand-confirmed (`B1`/DOM task 597) and scores `earned` here; a text heuristic for the pattern was tried and rejected because model self-report cannot separate deliberating from acting. Source: `reddit_sidebar_leakage_audit_with_wa.json`.

benchmark	cell · mode	scored successes	of which LEAKED	share
VWA	B0 · DOM	5	1	20.0%
VWA	B0 · SoM-image	2	0	0.0%
VWA	B0 · DOM+sprompt	3	0	0.0%
VWA	B0 · DOM+stext	3	0	0.0%
VWA	B0 · SoM	3	0	0.0%
VWA	B0 · Vision	2	1	50.0%
VWA	B1 · DOM	2	0	0.0%
VWA	B1 · SoM-image	4	0	0.0%
VWA	B1 · DOM+sprompt	2	0	0.0%
VWA	B1 · DOM+stext	4	0	0.0%
VWA	B1 · SoM	3	1	33.3%
VWA	B2 · DOM	3	3	100.0%
VWA	B2 · SoM	1	0	0.0%
WA	B0 · DOM	3	0	0.0%
WA	B0 · SoM-image	5	0	0.0%
WA	B0 · DOM+stext	5	0	0.0%
WA	B0 · SoM	3	0	0.0%
WA	B0 · Vision	3	0	0.0%
WA	B1 · DOM	4	0	0.0%
WA	B1 · SoM-image	5	0	0.0%
WA	B1 · DOM+stext	5	0	0.0%
WA	B1 · SoM	3	0	0.0%
WA	B1 · Vision	1	0	0.0%

Table 44: The six observation modes (§2). The three P-modes keep either the mark legend, the SoM prompt, or both, while removing the per-step screenshot.

mode	text payload	prompt family	annotated screenshot
DOM	accessibility tree	DOM	no
SoM	mark legend	SoM	**yes**
Vision	none	vision	**yes** (unannotated)
P-text	mark legend	DOM	no
P-prompt	accessibility tree	SoM	no
P-SoM	mark legend	SoM	no

successes that the fixed policy does not collect, against four collected by the fixed policy overall. The permutation null detects that enrichment. A globally ordered score is not required to produce it, which is why the cell's AUROC of 0.483, below both chance and its own best single covariate at 0.711, is consistent with a real saving.

Two properties of that test are worth recording. It runs 10,000 draws and reports the plus-one Monte Carlo estimator (k+1)/(B+1), whose floor is therefore 1.0×10^{-4}, two orders below the tightest Holm threshold of 8.3×10^{-3}. That matters because at the 200 draws we first used, the floor was $1/201 = 4.98 \times 10^{-3}$, this cell reported exactly it, and whether it could clear the threshold at all was a function of the draw count rather than of the data; at 10,000 draws four of the draws match or beat the observed saving, so $p = 5.0 \times 10^{-4}$ is measured. Second, the quantity tested is the saving at an operating point selected against whole-cell outcomes, which is not the nested policy of Table 6. Null and observation select that point the same way, so the null absorbs the selection optimism and the

Table 45: Per-cell detail behind §6. The last column counts labelled tasks on which the fixed priority list selected a mode strictly more expensive than another mode that also succeeded on that task. The exact-tie case, where the list order is the only tiebreaker, occurs in zero rows in every cell.

cell	labels	multi-success	list picked a strictly pricier mode
classifieds · B0	97	68 (70.1%)	**53 (54.6%)**
reddit · B0	53	36 (67.9%)	**23 (43.4%)**
classifieds · B1	55	29 (52.7%)	**26 (47.3%)**
reddit · B1	24	17 (70.8%)	**9 (37.5%)**
classifieds · B2	16	4 (25.0%)	**2 (12.5%)**
reddit · B2	15	3 (20.0%)	**2 (13.3%)**

Table 46: Per-cell detail behind §6. The naive column selects the threshold and the strongest and cheapest modes from whole-cell outcomes; the nested column re-derives all three inside each outer fold. Nesting moves the result in both directions.

cell	naive nesting SR	fully nested SR	Δ
classifieds · B0	25.45%	26.79%	**+1.34pp**
classifieds · B1	13.84%	14.29%	+0.45pp
classifieds · B2	1.34%	1.34%	±0.00pp
reddit · B0	13.79%	12.81%	−**0.99pp**
reddit · B1	6.40%	5.91%	−0.49pp
reddit · B2	3.94%	3.94%	±0.00pp

Table 47: Per-site detail behind Appendix L.3. A conflict is one task on which two cells recorded different oracle modes. In-sample modal agreement is the accuracy of emitting the modal label per task, over all pooled labelled rows (168 on classifieds, 92 on reddit), a row being one (task, backbone) pair on which that backbone solved the task.

site	tasks labelled in $\geq$2 cells	conflicting	conflict rate	in-sample modal agreement	same, on shared tasks only	same, exact-vector grouping
classifieds	54	31	**57.4%**	**79.2%**	70.3% (n=118)	83.9%
reddit	25	14	**56.0%**	**83.7%**	74.1% (n=58)	89.1%

comparison is fair, but the point is not one a deployment could occupy, and §6's conclusion rests on the nested numbers rather than on this test.

M.1 Supply and trainability under both label definitions

§6 keeps the prior-order label and reports the measured-cost rule as a sensitivity. Supply is identical under both by construction, since each labels exactly the tasks some mode solved, so only the class distribution and through it trainability can move. The relabelled column equals the "order picked a strictly pricier mode" column of A.2 exactly, which is the consistency check: a label moves if and only if the order's pick was not the measured cheapest.

The single cell that changes, classifieds · B1, loses a class rather than gaining one: the prior-order label keeps DOM and SoM above the threshold, the measured-cost label concentrates enough of those rows onto Vision that only Vision survives. The alternative definition therefore strengthens the negative result, which is a reason to report it and not a reason to adopt it.

M.2 The best-success mode is not stable across folds

The nested design of §6 re-selects the best-success mode inside every outer fold, which exposes something the whole-cell version conceals. In reddit · B0 the five outer folds select DOM, DOM, SoM, SoM, DOM. A pipeline that picks one best mode from all realised outcomes is therefore not

Table 48: Per-site detail behind Appendix L.4, over the same solve events. Columns two and three carry the same resubstitution caveat as Appendix K.4, and column three additionally rises for an arithmetic reason: merging six classes into two can only increase a modal share. The claim that backbones agree about the screenshot therefore rests on the last two columns, which measure agreement between two backbones' labels directly rather than against a modal label. Under Appendix K.4's exact-vector grouping the tier figures are 92.3% and 97.8%. No classifier is fitted to this target anywhere in the paper.

site	which-mode modal agreement	tier modal agreement	tier agreement across backbones	six-way agreement across backbones
classifieds	79.2%	**89.9%**	**68.5%**	42.6%
reddit	83.7%	**96.7%**	**88.0%**	44.0%

Table 49: Trainability under the two label definitions. "Surviving" counts classes clearing ten training rows in a five-fold split; **no** marks a cell with fewer than two, where no classifier exists. Four of six cells are untrainable under the reported label and five of six under the measured-cost alternative, so the supply argument does not depend on the choice.

cell	labels	relabelled	prior order: surviving classes	measured cost: surviving classes
classifieds · B0	97	53	3 (DOM, P-prompt, SoM)	2 (SoM, Vision)
reddit · B0	53	23	1 (DOM) (**no**)	1 (P-text) (**no**)
classifieds · B1	55	26	2 (DOM, SoM)	1 (Vision) (**no**)
reddit · B1	24	9	0 (**no**)	0 (**no**)
classifieds · B2	16	2	0 (**no**)	0 (**no**)
reddit · B2	15	2	0 (**no**)	0 (**no**)

merely optimistic about its threshold; it reports a mode choice that its own resampling does not reproduce.

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